

Microsoft Azure

1



Microsoft Azure is a cloud computing platform that helps you build, deploy and manage applications through Microsoft's global network.

① What is Azure?



- Microsoft's cloud platform.
- Offers 200+ products and cloud services.
- Build, deploy and manage applications globally.
- Pay-as-you-go pricing.

② Cloud Computing



- Delivers IT services over the internet.
- On-demand resources.
- No need to manage physical hardware.
- Elastic, scalable and reliable.

③ Azure Benefits

- 💰 Cost Effective - Pay-as-you-go model.
- 🕒 Scalability - Scale up or down easily.
- 🛡️ High Availability - Built-in redundancy.
- 🔒 Security - Enterprise-grade security.
- 🌐 Global Reach - 60+ regions worldwide.
- 🚀 Innovation - Access to latest technologies.
- 📊 Reliability - 99.95% SLA for most services.

④ Azure Global Infrastructure



- 60+ Azure Regions across the globe.
- Each Region has multiple Availability Zones.
- Provides high availability, performance and data residency.
- Regions are paired for disaster recovery.

- 📍 North America
- 📍 Europe
- 📍 Asia
- 📍 India
- 📍 Australia
- and more...

⑤ Popular Azure Services



Virtual Machines



Storage Account



Azure SQL Database



Azure Kubernetes Service (AKS)



Azure Functions



Azure Virtual Network



Microsoft Entra ID



Azure Monitor



Azure Key Vault



REMEMBER

- Azure is Microsoft's Cloud Platform.
- Resources are organized in a hierarchy.
- Everything in Azure is a resource.
- You pay only for what you use.



BEST PRACTICES

- Follow the Well-Architected Framework.
- Use Tags for all resources.
- Enable Monitoring and Alerts.
- Secure your resources and follow least privilege.
- Plan for High Availability and Backup.

Azure Global Infrastructure

2



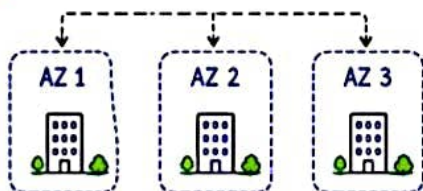
Azure provides a global network of data centers to build highly available, scalable and resilient applications.

1 Regions



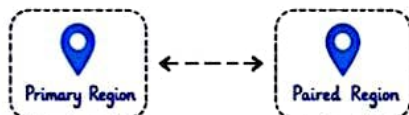
- A region is a geographical area that contains one or more data centers.
- Azure has 60+ regions worldwide.

2 Availability Zones



- Physically separate data centers within a region.
- Protects applications from data center failures.

3 Region Pair



- Each region is paired with another region hundreds of miles away.
- Used for geo-redundant storage and disaster recovery.

4 Edge Zones



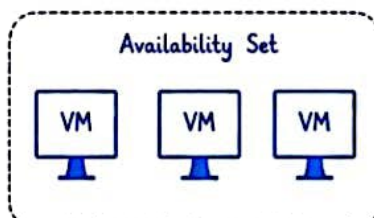
- Located closer to end users.
- Extends Azure services to edge of the network.
- Ideal for low-latency applications.



REMEMBER

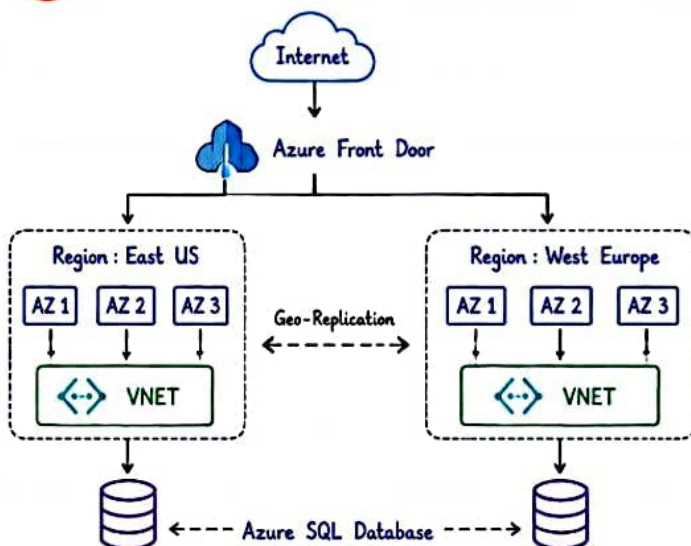
- Regions = Geographical areas.
- Zones = Data center level HA.
- Region Pair = DR & Geo-redundancy.
- Edge Zones = Low latency at the edge.
- Design for High Availability & Resilience.

5 Availability Sets



- Distributes VMs across fault domains and update domains.
- Protects against hardware failures and maintenance updates.

6 Real-world Example



7 Best Practices

- ✓ Choose the right region close to your users.
- ✓ Use Availability Zones for high availability.
- ✓ Use Region Pair for disaster recovery.
- ✓ Use Edge Zones for low-latency workloads.
- ✓ Design for resilience and fault tolerance.
- ✓ Monitor and test failover regularly.



BEST PRACTICES

- ✓ Use Availability Zones for production workloads.
- ✓ Enable Geo-redundant storage.
- ✓ Keep critical data and apps in paired regions.
- ✓ Use Edge Zones for real-time applications.
- ✓ Regularly review and test your DR strategy.



CLOUD
ARCHITECTURE



DEVOPS
AUTOMATION



SECURITY &
COMPLIANCE



COST OPTIMIZATION
& FINOPS

DESIGN

OPTIMIZE

SECURE

SCALE

Azure Identity

Secure access & identity for users, apps and services across Azure and beyond.



3

① Microsoft Entra ID (Azure AD)



- ✓ Cloud-based identity and access management.
- ✓ Successor of Azure AD.
- ✓ Authenticate and authorize users, apps and services.
- ✓ Supports hybrid & multi-cloud.

② Users



- ✓ Represent people in Entra ID.
- ✓ Can access Azure resources.
- ✓ Assigned to groups & roles.
- ✓ Supports external users (guest users).

③ Groups



- ✓ Collection of users.
- ✓ Simplifies access management.
- ✓ Used for Role Assignment.
- ✓ Types: Security, Microsoft 365.

④ Service Principal



- ✓ Identity for applications.
- ✓ Used by apps, automation and CI/CD pipelines.
- ✓ Supports RBAC and access control.

⑤ Managed Identity



- ✓ Managed identity for Azure resources.
- ✓ No credential management.
- ✓ Used by VMs, AKS, Functions.
- ✓ More secure than secrets.

⑥ RBAC (Role Based Access Control)



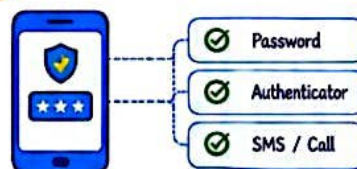
- ✓ Control access to Azure resources.
- ✓ Assign roles to Users, Groups, Service Principals or Managed Identity.
- ✓ Principle of Least Privilege.
- ✓ Built-in, Custom and Azure AD roles.

⑦ Conditional Access



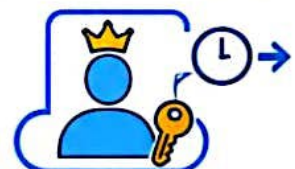
- Apply access policies.
- Based on user, device, location.
- Increases security posture.

⑧ MFA (Multi-Factor Authentication)

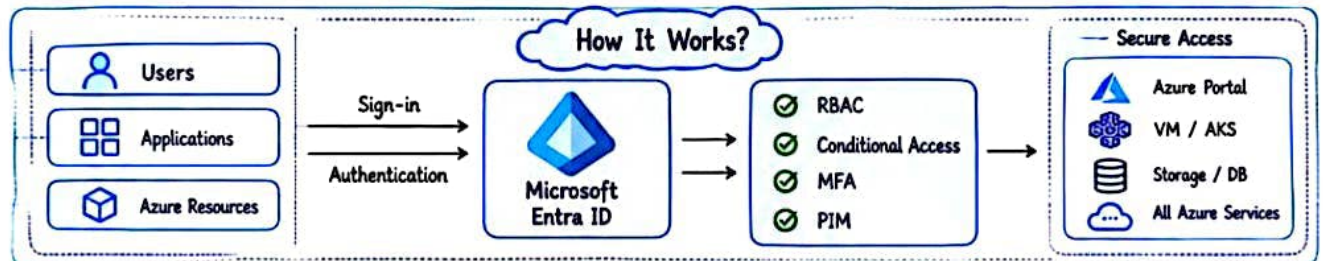


- Adds extra layer of security.
- Reduces risk of account compromise.
- Recommended for all users.

⑨ PIM (Privileged Identity Management)



- Just-in-Time (JIT) access to roles.
- Time-bound & approval based.
- Reduces standing admin access.
- Enhances security & compliance.



REMEMBER

- ✓ Entra ID = Single Identity Platform for Microsoft Cloud.
- ✓ Use Groups for simpler access management.
- ✓ Use Managed Identity for workloads (no secrets).
- ✓ RBAC ensures controlled access to resources.
- ✓ MFA & Conditional Access are must for security.



BEST PRACTICES

- ✓ Follow Least Privilege access model.
- ✓ Use MFA for all users and admins.
- ✓ Enable PIM for privileged roles.
- ✓ Automate access with Managed Identity & Service Principal.
- ✓ Regularly review roles and access (Access Reviews).



CLOUD ARCHITECTURE



DEVOPS AUTOMATION



SECURITY & COMPLIANCE



COST OPTIMIZATION & FINOPS

DESIGN

OPTIMIZE

SECURE

SCALE

Azure Virtual Machines

Azure Virtual Machines provide scalable, on-demand computing resources in the cloud.



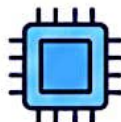
4

① VM Basics



- On-demand scalable servers.
- Supports Windows & Linux.
- Complete control of OS.
- Integrates with Azure services.

② VM Sizes



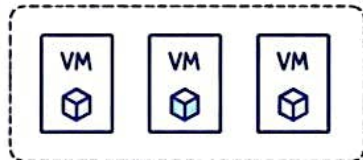
- Wide range of sizes for different workloads.
- General Purpose (D)
- Compute Optimized (F)
- Memory Optimized (E)
- GPU Optimized (NC)
- Storage Optimized (L, M)

③ Managed Disks



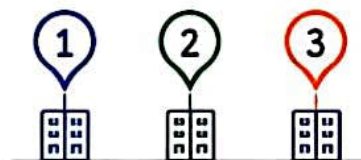
- Durable & highly available storage.
- Types: Standard HDD, Standard SSD, Premium SSD
- Auto-redundant storage.

④ Availability Set



- Distributes VMs across fault domains and update domains.
- Protects against hardware failures and maintenance events.

⑤ Availability Zones



- Physically separate data centers within a region.
- Protects against datacenter failures.
- Higher availability and resilience.

⑥ Scale Set



- Automatically scales VMs in or out.
- Based on demand or schedule.
- Load balanced and highly available.
- Reduces cost and improves availability.

⑦ VM Deployment Flow



- User requests VM → Azure validates → Resources created → VM running.
- Takes few minutes.

⑧ Bastion

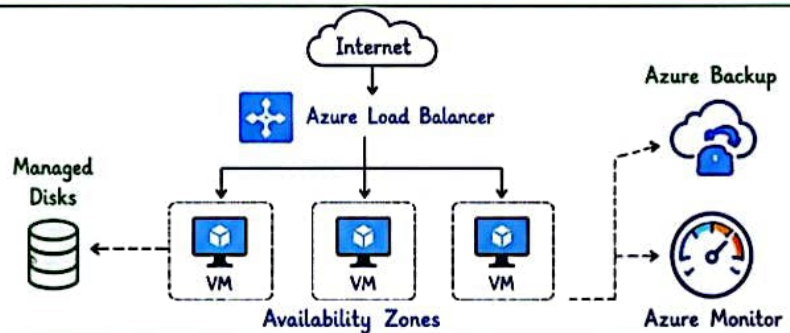


- Secure RDP/SSH access to VM over SSL.
- No public IP required on VM.
- Protects VM from internet exposure.

⑨ Backup



- Azure Backup protects VM and its data.
- Supports full & incremental backup.
- Restore to same region or cross region.
- Retention policy configurable.



REMEMBER

- VMs are the building blocks of compute in Azure.
- Use Managed Disks for high availability and durability.
- Use Availability Zones for highest resiliency.
- Use Scale Set for auto scaling and high availability.
- Use Bastion for secure access without public IP.
- Always enable Backup for important workloads.



CLOUD
ARCHITECTURE



DEVOPS
AUTOMATION



SECURITY &
COMPLIANCE



COST OPTIMIZATION
& FINOPS

DESIGN | OPTIMIZE | SECURE | SCALE

Azure Storage

Azure Storage provides durable, highly available, scalable and secure data storage for modern cloud applications.



5

1 Blob Storage



- Stores unstructured data as objects.
- Ideal for images, videos, backups, logs, etc.
- Supports Hot, Cool, Archive access tiers.

2 File Storage



- Fully managed file shares in the cloud.
- Access via SMB or NFS.
- Lift & shift applications.
- Used by VMs and on-prem apps.

3 Queue Storage



- Stores messages for asynchronous processing.
- First-in, first-out model.
- Decouples application components.

4 Table Storage



- NoSQL key/value store.
- Schema-less and highly scalable.
- Stores large amounts of structured data.

5 Managed Disk



- Block-level storage for Azure VMs.
- Types: Standard HDD, Standard SSD, Premium SSD.
- High performance and durability.

6 Storage Account



- Container for all storage data objects.
- Unique namespace across Azure.
- Define redundancy, performance & access.

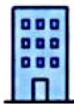
7 Access Tier



- Hot - Frequent access, higher cost.
- Cool - Infrequent access, lower cost.
- Archive - Rare access, lowest cost.

8 LRS

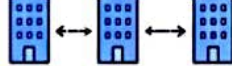
(Locally Redundant Storage)



- 3 copies within the same data center.
- Low cost.
- Protects against disk failures.

9 ZRS

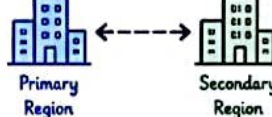
(Zone Redundant Storage)



- 3 copies across Availability Zones.
- High availability within a region.
- Protects against zone failures.

10 GRS

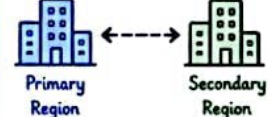
(Geo Redundant Storage)



- 3 copies in primary region.
- 3 copies in a secondary region hundreds of miles away.
- Protects against regional failures.

11 RA-GRS

(Read-Access Geo Redundant Storage)



- Same as GRS with read access to secondary region.
- Enables business continuity and DR scenarios.

12 Lifecycle Management



Blob Data



Define Policy



Policy Execution



- Automatically move data across access tiers based on rules.
- Optimize cost by moving less accessed data to Cool or Archive.
- Policies based on days since last access.



Best Practices

- ✓ Use the right storage type for your workload.
- ✓ Enable redundancy based on availability requirements.
- ✓ Use lifecycle management to optimize storage cost.
- ✓ Secure data using Azure RBAC, SAS & Encryption.
- ✓ Monitor storage usage and set up alerts.
- ✓ Use Private Endpoints for secure access.
- ✓ Regularly review and clean up unused data.



REMEMBER

- Blob = Unstructured Data
- File = Shared File System (SMB/NFS)
- Queue = Messaging
- Table = NoSQL Data
- Choose the right redundancy for durability & availability.
- Use lifecycle management to reduce storage cost.

Redundancy	Copies	Scope	Best For
LRS	3	Single Data Center	Low cost, non-critical data
ZRS	3	Across Availability Zones	High availability
GRS	6	Geo-redundant (Read-Only)	DR, backup, critical data
RA-GRS	6	Geo-redundant (Read/Write)	DR with read access



CLOUD
ARCHITECTURE



DEVOPS
AUTOMATION



SECURITY &
COMPLIANCE



COST OPTIMIZATION
& FINOPS

DESIGN | OPTIMIZE | SECURE | SCALE

Azure Networking

Azure Networking enables secure, scalable and highly available communication between resources, users and the internet.



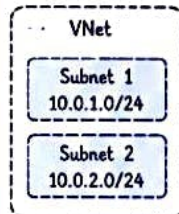
6

① Virtual Network (VNet)



- Foundation of Azure network.
- Isolated network in Azure cloud.
- Define your own IP address space.
- Enables communication between resources.

② Subnet



- Subnet is a range of IP addresses in a VNet.
- Enables segmentation of resources.
- Each subnet resides in a single region.

③ NSG (Network Security Group)



- Controls inbound and outbound traffic.
- Rule based allow/deny traffic.
- Associated to subnet or NIC.
- Stateful and redundant.

④ ASG (Application Security Group)



- Logical group of VMs with similar function.
- Simplifies NSG rule management.
- Use in place of IP addresses.

⑤ Route Table



- Contains a set of routes that determine where network traffic is forwarded.
- Associated with subnet.

⑥ UDR (User Defined Route)

Prefix	Next Hop Type
10.1.0.0/16	VNet
0.0.0.0/0	NVA
172.16.0.0/16	VPN Gateway

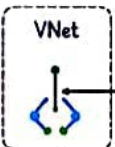
- Custom routes defined by user.
- Override system routes.
- Enable advanced traffic control.

⑦ Public IP



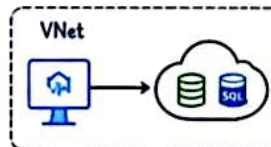
- Publicly reachable IP on the internet.
- Used by resources like VM, Load Balancer.
- Static or Dynamic.

⑧ Private Endpoint



- Private connectivity to Azure services.
- Uses Private Link.
- Traffic stays on Microsoft backbone.

⑨ Service Endpoint



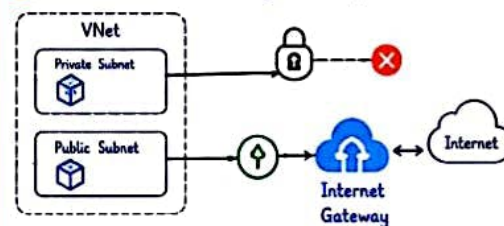
- Extends VNet to Azure services.
- Uses Azure backbone network.
- Easier to configure than Private Endpoint.

⑩ NAT Gateway



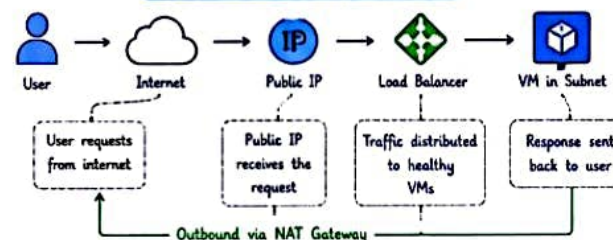
- Enables outbound internet access for resources in private subnets.
- Uses static public IP(s).
- Highly available, scalable and secure.

⑪ Internet Gateway Concept



- Internet Gateway allows communication between VNet and internet.
- Resources in Public Subnet are accessible from internet.
- Private Subnet resources are not directly accessible.

How Traffic Flows in Azure



★ Best Practices

- ✓ Use VNets to isolate and secure your resources.
- ✓ Use NSGs to control and restrict traffic.
- ✓ Use ASGs for easier security rule management.
- ✓ Use UDR for custom routing requirements.
- ✓ Use Private Endpoints for secure access to PaaS services.
- ✓ Use NAT Gateway for secure outbound internet access.
- ✓ Enable Azure Firewall for advanced threat protection.
- ✓ Regularly review NSG rules and route tables.

💡 REMEMBER

- VNet is the foundation of all Azure networking.
- Subnets help in network segmentation.
- NSG controls traffic, Route Table controls path.
- Private Endpoint keeps traffic on Microsoft network.

Replication Type	Copies	Where	Best For
LRS (Locally Redundant Storage)	3	Same Data Center	Low cost, non-critical data
ZRS (Zone Redundant Storage)	3	Across Availability Zones	High availability
GRS (Geo Redundant Storage)	6	Primary + Secondary Region	DR, backup, critical data
RA-GRS (Read-Access Geo Redundant Storage)	6	Primary (R/W) + Secondary (R)	DR with read access

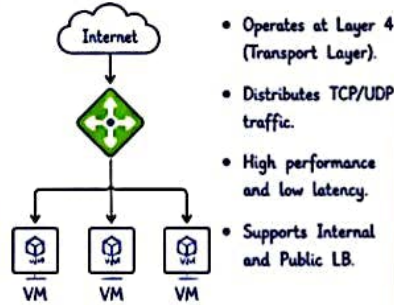
Azure Load Balancer

Azure Load Balancer services help distribute traffic, improve performance and ensure high availability of applications.

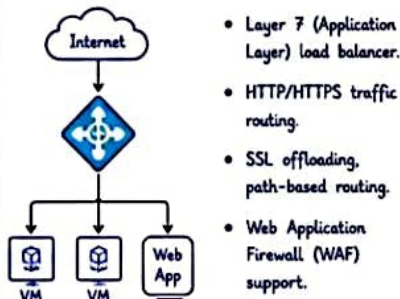


7

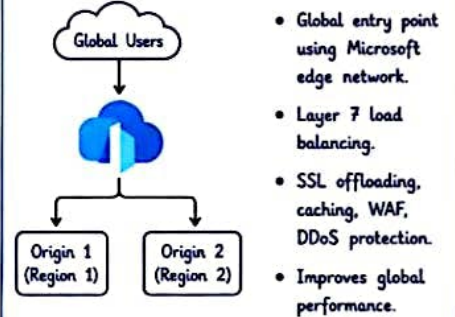
① Load Balancer



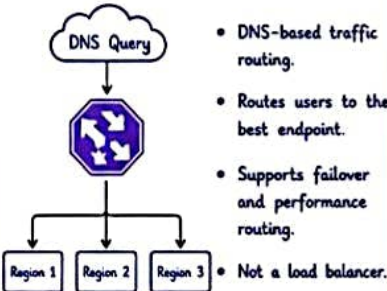
② Application Gateway



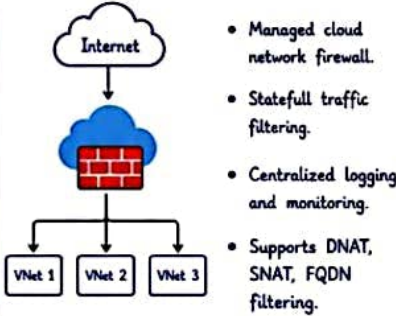
③ Azure Front Door



④ Traffic Manager



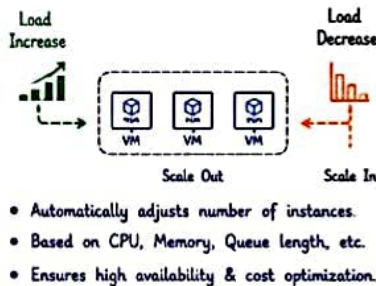
⑤ Azure Firewall



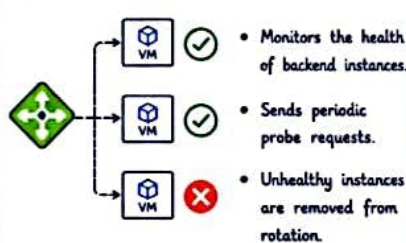
⑥ WAF (Web Application Firewall)



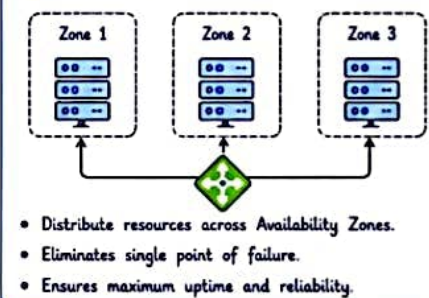
⑦ Autoscale



⑧ Health Probe



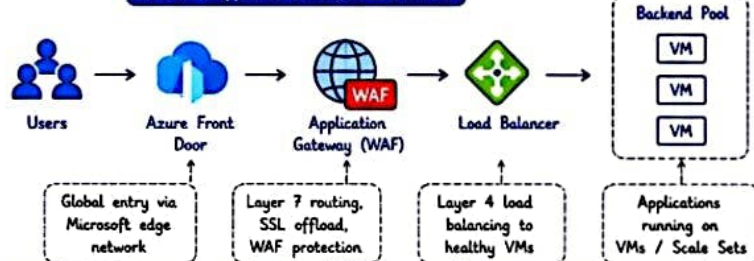
⑨ High Availability



★ ⑩ Best Practices

- Use the right service for the right scenario.
- Enable Availability Zones for high availability.
- Use Health Probes to detect unhealthy instances.
- Enable WAF to protect web applications.
- Use Autoscale to handle traffic variations.
- Use Azure Front Door for global applications.
- Monitor metrics and set up alerts.
- Keep backend resources updated and secure.

How Traffic is Routed in Azure



REMEMBER

- Load Balancer = Layer 4 (TCP/UDP)
- Application Gateway = Layer 7 (HTTP/HTTPS)
- Front Door = Global Layer 7 with Edge network
- Traffic Manager = DNS-based routing
- Use Health Probe + Autoscale for resiliency

Service	Layer	Type	Use Case
Load Balancer	4	Transport (TCP/UDP)	High performance, simple load balancing
Application Gateway	7	Application (HTTP/HTTPS)	Web apps, routing, SSL, WAF
Azure Front Door	7	Global Edge	Global apps, caching, WAF, performance
Traffic Manager	DNS	DNS-based	Global routing, failover
Azure Firewall	3-7	Firewall	Network security and filtering

Azure DNS

8



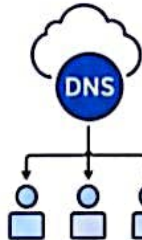
Azure DNS is a scalable, reliable and cost-effective DNS hosting service that provides name resolution using Microsoft's global network.

① DNS (Domain Name System)



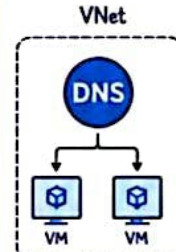
- Translates domain names to IP addresses.
- Essential for all internet communication.
- Uses a hierarchical and distributed system.

② Public DNS Zone



- Resolves names for public internet.
- Hosted in Azure DNS service.
- Accessible from anywhere on internet.
- Example: example.com

③ Private DNS Zone

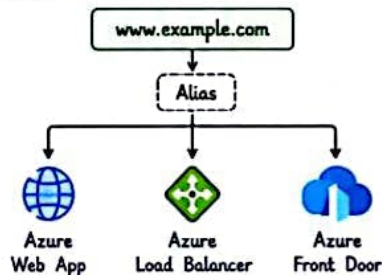


- Resolves names within virtual networks.
- Not accessible over the internet.
- Linked to one or more VNets.
- Example: contoso.local

④ Record Types

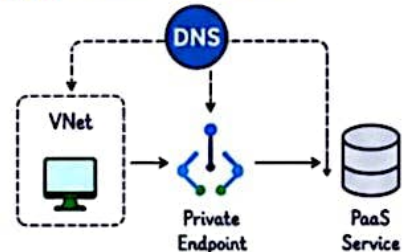
Type	Description
A	Maps domain to IPv4 address
AAAA	Maps domain to IPv6 address
CNAME	Maps domain to another domain
MX	Mail exchange record
TXT	Text information
NS	Name server record
SRV	Service location record

⑤ Alias Record



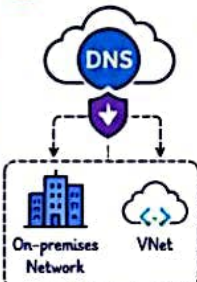
- Maps a DNS name to Azure resource.
- Automatically updates with IP changes.
- Supports zone apex (root domain).

⑥ Private Link DNS



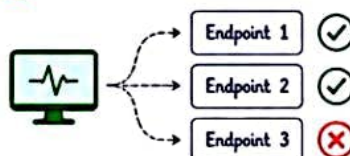
- Resolves Private Endpoint names.
- Azure manages Private DNS integration.
- Ensures private and secure connectivity.

⑦ Resolver (Azure DNS Resolver)



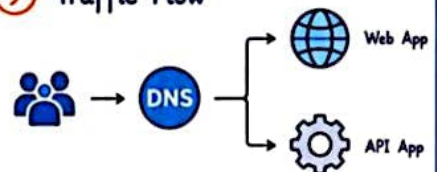
- Provides DNS resolution between on-premises and Azure.
- Uses Inbound & Outbound Endpoints.
- Forward DNS queries to on-prem DNS.
- Supports hybrid cloud environments.

⑧ Health Check



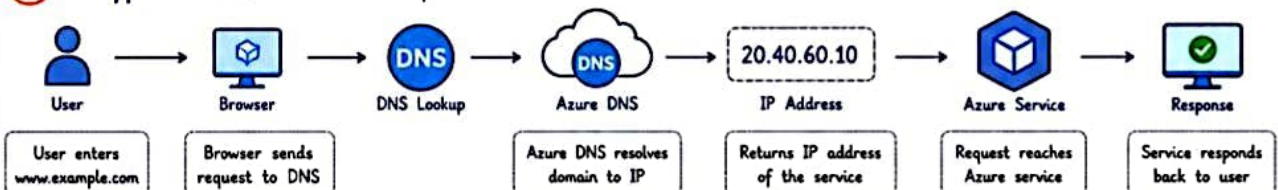
- Monitors the health of endpoints.
- Uses HTTP, HTTPS or TCP checks.
- Helps in traffic routing and failover.
- Used by Traffic Manager.

⑨ Traffic Flow



- User requests a domain name.
- DNS resolves name to IP address.
- Traffic is routed to the appropriate Azure service.

⑩ Traffic Flow (End-to-End Example)



REMEMBER

- DNS is the phonebook of the internet.
- Use Private DNS Zone for internal name resolution.
- Use Alias Record for Azure resources.
- Use Azure DNS Resolver for hybrid environments.
- Health Checks improve reliability with Traffic Manager.



BEST PRACTICES

- ✓ Use appropriate DNS zones (Public or Private).
- ✓ Enable Alias Record for Azure resources.
- ✓ Use Private DNS with Private Endpoints.
- ✓ Implement DNS Resolver for hybrid scenarios.
- ✓ Monitor DNS with Azure Monitor.
- ✓ Use Health Checks with Traffic Manager for high availability.



CLOUD
ARCHITECTURE



DEVOPS
AUTOMATION



SECURITY &
COMPLIANCE



COST OPTIMIZATION
& FINOPS

DESIGN | OPTIMIZE | SECURE | SCALE

Azure App Service

9



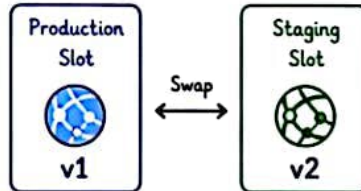
Azure App Service is a fully managed platform for building, deploying and scaling web, mobile and API applications in the cloud.

1 Web Apps



- Host web applications, APIs and mobile backends.
- Supports .NET, Java, Node.js, Python, PHP, Go, etc.
- Automatic scaling and built-in load balancing.
- Integrated with Azure services.

2 Deployment Slots



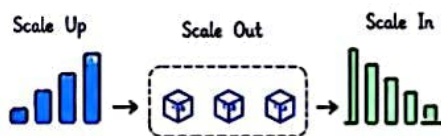
- Create multiple slots in the same App.
- Test changes in Staging/Demo slot.
- Swap with Production slot in one click.
- Zero-downtime deployments.

3 App Service Plan



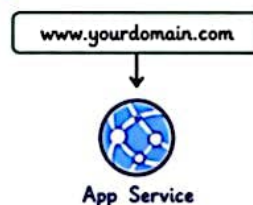
- Defines compute resources for your app.
- Pricing tiers:
 - Free
 - Shared
 - Basic
 - Standard
 - Premium (V2/V3)
- All apps in a plan share the resources.

4 Scaling



- Scale Up - Increase size (CPU/RAM).
- Scale Out - Add more instances.
- Scale In - Reduce instances to save cost.
- Supports manual and auto scaling rules.

5 Custom Domain



- Map your custom domain to App Service.
- Update DNS with CNAME record.
- Supports apex domain and subdomain.

6 SSL (HTTPS)



https://

- Enable HTTPS for secure access.
- Use Free SSL (App Service Managed Certificate).
- Or integrate with Azure Key Vault for custom SSL.
- Auto-renewal supported.

7 Monitoring



- Built-in Application Insights.
- Real-time logs and metrics.
- Performance monitoring.
- Custom alerts and dashboards.
- Log Analytics integration.



Metrics



Logs



Alerts



Dashboards



App Insights

8 Deployment Options



GitHub Actions



Azure DevOps



Git



FTP / FTPS



ZIP Deploy



Docker Container



Code



Build



Test



Deploy



App Service

CI/CD pipeline for continuous delivery



REMEMBER

- App Service is a PaaS offering.
- It abstracts infrastructure management.
- Apps run in App Service Plan.
- Supports multiple languages and frameworks.
- Focus on code, not on servers.



BEST PRACTICES

- ✓ Use appropriate App Service Plan for your workload.
- ✓ Enable HTTPS and enforce secure connections.
- ✓ Use Deployment Slots for safe releases.
- ✓ Enable Auto Scaling for scalability and cost optimization.
- ✓ Monitor with Application Insights and set alerts.
- ✓ Use Managed Identity for secure access to Azure resources.
- ✓ Regularly review logs, metrics and performance.



CLOUD ARCHITECTURE



DEVOPS AUTOMATION



SECURITY & COMPLIANCE



COST OPTIMIZATION & FINOPS

DESIGN

OPTIMIZE

SECURE

SCALE

Azure Kubernetes Service (AKS)

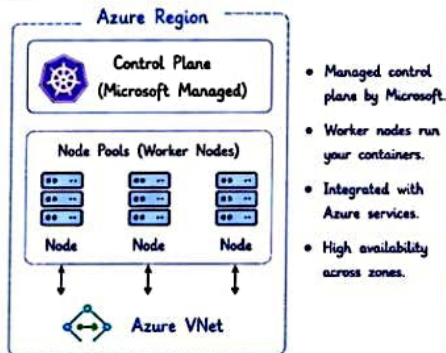
10



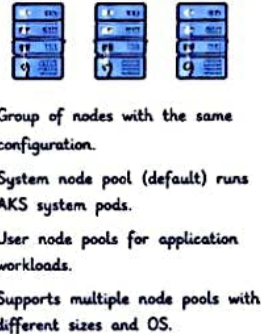
Azure Kubernetes Service (AKS)

AKS is a fully managed Kubernetes service on Azure that makes it easy to deploy, manage and scale containerized applications.

1 AKS Architecture



2 Node Pool

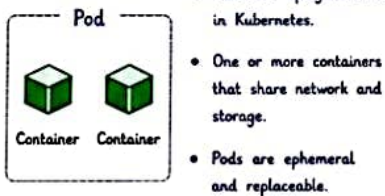


3 Control Plane

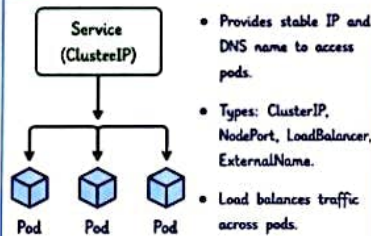


- Fully managed by Azure.
- Handles scheduling, API server, etcd, controller manager.
- Highly available and automatically upgraded.
- Not accessible directly from the internet (private by default).

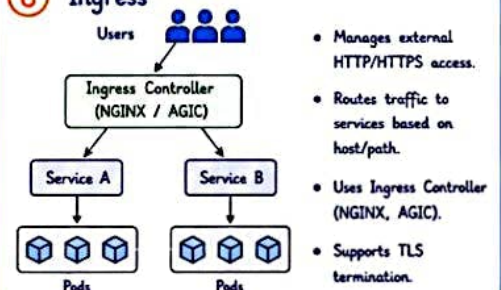
4 Pods



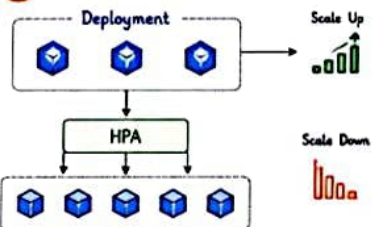
5 Service



6 Ingress

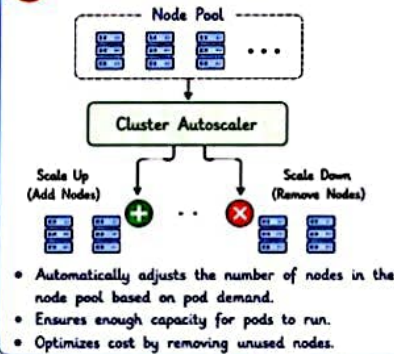


7 HPA (Horizontal Pod Autoscaler)

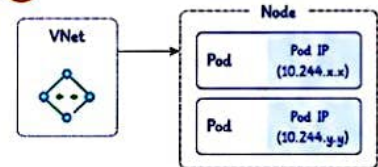


- Automatically scales pods based on CPU/Memory or custom metrics.
- Maintains application performance and reduces cost.
- Works with metrics server.

8 Cluster Autoscaler

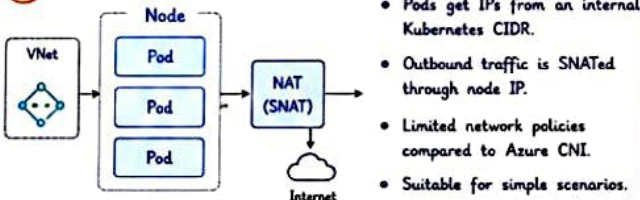


9 Azure CNI



- Each pod gets an IP address from the Azure VNet.
- Uses Azure VNet IP address space.
- Better network performance and Azure network policy support.

10 Kubenet



11 Key AKS Networking Options Comparison

Feature	Azure CNI	Kubenet
Pod IP	Azure VNet IP	Kubernetes CIDR
Network Policy	Azure Network Policy	Limited (Calico only)
Performance	Higher	Lower
SNAT	Not required	Required (Node SNAT)
Best For	Production workloads	Simple / Dev workloads

REMEMBER

- AKS = Managed Kubernetes on Azure.
- Control Plane is managed by Microsoft.
- Pods are ephemeral; use Deployments for resilience.
- Services provide stable access to pods.
- Ingress exposes HTTP/HTTPS applications.
- HPA scales pods; Cluster Autoscaler scales nodes.
- Azure CNI is recommended for production.

BEST PRACTICES

- Use multiple node pools for different workload types.
- Enable Auto Upgrades and Node Image Auto Patching.
- Use Azure Monitor and Container Insights for observability.
- Enable Network Policy (Azure CNI) for security.
- Use Managed Identities for pod-to-Azure resource access.
- Store secrets in Azure Key Vault using CSI Driver.
- Regularly review RBAC roles and cluster access.
- Plan for backups and DR (Velero, etc.).

Azure DevOps

Azure DevOps is a suite of development tools and services that helps teams plan work, collaborate on code development, and build & deploy applications.



11

① Azure Repos



- Unlimited Git repositories (private & public).
- Branch policies and pull requests.
- Code review and collaboration.

② Boards



- Plan, track and discuss work.
- Kanban boards, backlogs, sprints.
- Work items: User Story, Bug, Task, Epic.

③ Pipelines



- CI/CD automation.
- Build, test and deploy code.
- Supports YAML and Classic pipelines.

④ Artifacts



- Create, host and share packages.
- Supports NuGet, npm, Maven, Python, etc.
- Universal Packages.

⑤ Test Plans



- Plan and track testing activities.
- Manual and exploratory testing.
- Test cases, test suites, test runs.

⑥ YAML Pipeline

pipeline.yaml (Example)

```
trigger:
- main

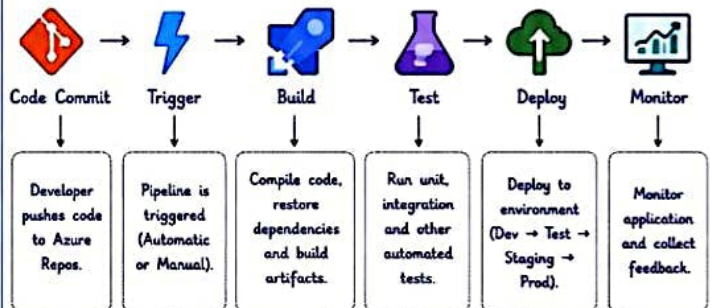
pool:
vmImage: 'ubuntu-latest'

stages:
- stage: Build
  jobs:
  - job: BuildJob
    steps:
    - script: echo "Building..."
      displayName: 'Build App'
- stage: Test
  dependsOn: Build
  jobs:
  - job: TestJob
    steps:
    - script: echo "Running Tests..."
      displayName: 'Test App'
- stage: Deploy
  dependsOn: Test
  jobs:
  - job: DeployJob
    steps:
    - script: echo "Deploying..."
      displayName: 'Deploy App'
```

Key Features

- Code as configuration (Everything in YAML).
- Version controlled and reusable.
- Multi-stage and multi-job pipelines.
- Secure with approvals and checks.
- Easily extend with tasks and templates.

⑦ CI/CD Flow



Benefits

- Faster Delivery
- High Quality
- Automation
- Consistency
- Visibility

⑧ Multi-stage Pipeline

Stages	Stage	Purpose	Approval	Example Environment
Build	Build	Compile and package application	✓	Build Agent
Test	Test	Run automated tests	✓	Test Environment
Staging	Staging	Deploy for UAT and validation	Optional	Staging Environment
Production	Production	Deploy to live users	Required	Production Environment

Use approvals, checks and gates to ensure quality and security.

⑨ Pipeline Components

Trigger	Defines when pipeline runs (CI/PR/Schedule/Manual).
Agent / Pool	Machine that runs the pipeline (Microsoft-hosted or Self-hosted).
Stages	Logical division of the pipeline (Build, Test, Deploy).
Jobs	A set of steps that run on the same agent.
Steps	Individual tasks or scripts executed in order.
Tasks	Predefined or custom actions (e.g. AzureCLI, Docker).
Artifacts	Files produced by a pipeline and used in later stages.
Variables	Store values used in pipeline (can be secret).



REMEMBER

- Azure DevOps is a complete ALM solution.
- Use Git branching policy and PR for better collaboration.
- Automate builds, tests and deployments with Pipelines.
- Store and share packages securely with Artifacts.
- Track work and collaborate using Boards.
- Always use approvals and checks for production deployments.



BEST PRACTICES

- ✓ Use YAML pipelines for version control and reusability.
- ✓ Implement branch policies and code reviews.
- ✓ Use variable groups and Key Vault for secrets.
- ✓ Run tests early and often.
- ✓ Use multi-stage pipelines with approvals for production.
- ✓ Monitor pipeline health and set up notifications.

Azure Monitoring

Azure Monitoring provides a comprehensive set of services to collect, analyze and act on telemetry data from your Azure resources and applications.



12

① Azure Monitor



- Centralized monitoring service for Azure.
- Collects metrics, logs and traces from resources.
- Provides insights, alerts and visualizations.
- Supports proactive and reactive monitoring.

② Log Analytics



- Log data platform in Azure Monitor.
- Stores and queries logs using KQL.
- Perform ad-hoc queries and analysis.
- Used for troubleshooting, compliance and reporting.

③ Application Insights



- Monitor web apps, mobile apps and APIs.
- End-to-end request tracing.
- Performance, availability and failure analysis.
- Real-time telemetry and user experience monitoring.

④ Alerts



- Get notified on important conditions.
- Alert-roles on metrics, logs and activity logs.
- Multiple action groups: Email, SMS, Webhook, ITSM, Azure Function.
- Supports smart detection and dynamic thresholds.

⑤ Metrics



- Numeric values that describe a resource.
- Platform, custom and diagnostic metrics.
- Real-time and historical metrics.
- Visualize with charts and set alerts.

⑥ Diagnostic Settings



- Send resource logs and metrics to destinations.
- Destinations: Log Analytics, Storage Account, Event Hub, Partner Solution.
- Enables centralized collection and auditing.

⑦ Workbook



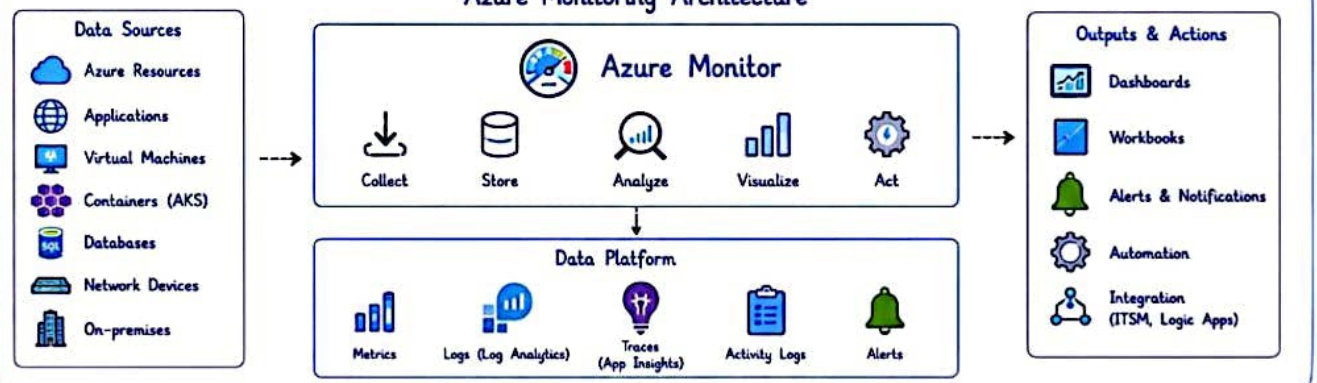
- Interactive reports with rich visualizations.
- Combine text, queries, charts and parameters.
- Shareable and customizable.
- Ideal for dashboards, reports and insights.

⑧ Dashboard



- Custom dashboards with tiles and charts.
- Drag and drop widgets.
- Share and collaborate.
- Real-time overview of your environment.

Azure Monitoring Architecture



① Azure Monitor



- Unified view of monitoring data.
- Supports metrics, logs, traces, activity logs.
- Built-in solutions for common scenarios.

② Log Analytics



- Kusto Query Language (KQL).
- Fast, powerful and scalable queries.
- Retention policies to manage cost.

③ Application Insights



- Tracks requests, dependencies, exceptions and performance.
- Supports distributed tracing.
- Built-in dashboards.

④ Alerts



- Static, Dynamic and Smart alerts.
- Suppression and Maintenance windows.
- Action Groups for notifications.

⑤ Metrics



- Monitor resource performance.
- Aggregation: Average, Sum, Min, Max.
- Granularity: 1 min to 1 day.
- Export metrics via API.

⑥ Diagnostic Settings



- Logs, Metrics and Activity Logs.
- Helps for compliance and auditing.
- Route data to multiple destinations.

⑦ Workbook



- Parameterized and interactive.
- Supports KQL, metrics and text.
- Export to PDF or Share.

⑧ Dashboard



- Pin any metrics, logs or resources.
- Easy to customize and share.
- Real-time operational overview.

REMEMBER

- Monitoring helps detect issues before they impact users.
- Use Log Analytics for deep dive and custom queries.
- Application Insights for application performance monitoring.
- Enable Diagnostic Settings for all critical resources.
- Alerts reduce MTTR (Mean Time To Resolve).

BEST PRACTICES

- Enable Azure Monitor for all subscriptions.
- Centralize logs in Log Analytics Workspace.
- Use Application Insights for user-facing applications.
- Create meaningful dashboards and workbooks.
- Set alerts for critical metrics and logs.
- Review and optimize retention to control costs.

Common Use Cases

- Performance Monitoring
- Availability Monitoring
- Security Monitoring
- Capacity Planning
- Cost Optimization

Azure Security

13



Azure provides a comprehensive set of security services to protect your cloud resources, applications, and data across identities, infrastructure and workloads.

1 Microsoft Defender for Cloud



- Unified security management and advanced threat protection.
- Continuous assessment and secure score.
- Workload protection for servers, containers, databases, storage, and more.
- Recommendations and regulatory compliance.

2 Microsoft Sentinel



- Cloud-native SIEM and SOAR solution.
- Collects data at scale from all sources.
- Detects threats with AI and built-in analytics.
- Automates response using playbooks.
- Incident investigation and threat hunting.

3 Key Vault



- Securely store and manage secrets, keys and certificates.
- Centralized key management.
- Integrates with Azure services and applications.
- Supports RBAC and access policies.

4 Azure Policy



- Create, assign and manage policies for resources.
- Enforce organizational standards and compliance.
- Audit or deny non-compliant resources.
- Built-in policies and custom definitions.

5 RBAC (Role-Based Access Control)



- Manage access to Azure resources.
- Built-in roles (Owner, Contributor, Reader, etc.).
- Assign roles at different scopes (MG, Subscription, RG, Resource).
- Follows the principle of least privilege.

6 NSG (Network Security Group)



- Controls inbound and outbound traffic to resources.
- Rule-based filtering (Allow/Deny).
- Applied to subnets or NICs.
- Stateful traffic filtering.
- Supports service tags and application security rules.

7 Firewall



- Network security service that protects resources.
- Stateful inspection and threat intelligence.
- Filter traffic by FQDN, IP, ports, and protocols.
- Centralized logging and monitoring.

8 WAF (Web Application Firewall)



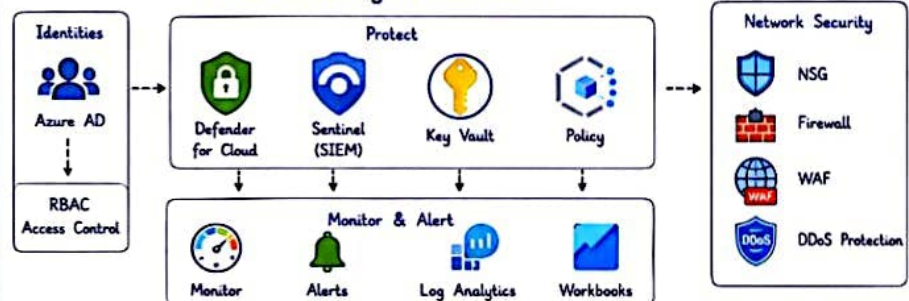
- Protects web applications from common attacks.
- OWASP Top 10 protection.
- SQL Injection, XSS, CSRF, RFI, etc.
- Integrated with Application Gateway, Front Door, App Service.

9 DDoS Protection



- Protects against DDoS attacks on your apps.
- Standard and Basic protection plans.
- Always-on traffic monitoring.
- Works with Azure Front Door, App GW, Public IP, and more.

Azure Security Architecture



11 Log Analytics



- Centralized log management and analysis.
- Collects logs from Azure resources and on-premises.
- KQL (Kusto Query Language) for powerful queries.
- Integrates with Sentinel, Monitor, and other services.

12 Application Insights



- Monitor and troubleshoot application performance.
- Real-time telemetry and usage analytics.
- Detect performance issues and failures.
- Supports .NET, Java, Node.js, Python, and more.

REMEMBER

- Security is a shared responsibility.
- Use least privilege access (RBAC).
- Enable MFA for all users.
- Encrypt data at rest and in transit.
- Monitor, alert, and respond to threats.
- Keep systems and dependencies up to date.
- Regularly review policies and access.
- Use built-in security services and automation.



BEST PRACTICES

- ✓ Enable Microsoft Defender for Cloud on all subscriptions.
- ✓ Use Microsoft Sentinel for centralized threat detection and response.
- ✓ Store secrets and keys in Key Vault - never in code.
- ✓ Enforce Azure Policy for compliance and governance.
- ✓ Apply RBAC with least privilege access.
- ✓ Restrict network access using NSG and Firewall.
- ✓ Protect web applications using WAF.
- ✓ Enable DDoS Protection for public-facing applications.
- ✓ Continuously monitor logs, metrics, and set up alerts.
- ✓ Conduct regular security assessments and reviews.



Common Threats & Protection

Threat	Protection
Unauthorized Access	RBAC, Conditional Access, MFA
Network Attacks	NSG, Firewall, DDoS Protection
Web Attacks	WAF, Application Gateway
Data Breach / Leakage	Key Vault, Encryption, DLP
Malware / Vulnerabilities	Defender for Cloud, Patching
Advanced Threats	Sentinel, Threat Intelligence

Azure Backup & Disaster Recovery

Protect your data, applications and infrastructure with Azure native backup and disaster recovery services.



14

① Azure Backup



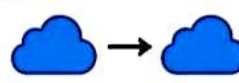
- Simple & secure backup for Azure workloads.
- Supports VM, SQL DB, Files, SharePoint, SAP HANA and more.
- Policy based backups with on-demand restore.
- Long term retention with Azure storage.

② Recovery Services Vault



- Centralized vault to manage backup and recovery.
- Store backup data securely in geo-redundant storage.
- Supports encryption, access control and soft delete.
- Single pane of glass for monitoring and reporting.

③ Site Recovery



- Replicate VMs to another Azure region.
- Minimize downtime with planned or unplanned failover.
- Supports test failover without impacting production.
- Orchestrated recovery with runbooks.

④ Cross Region Restore



- Restore backups to alternate Azure regions.
- Helps for region outage or permanent data loss.
- Improve business continuity and data availability.
- Supports GRS/GZRS storage for resiliency.

⑤ Soft Delete



- Protects backup data from accidental deletion.
- Retain recovery points for a specified period.
- Enhances data safety and compliance.

⑥ Immutable Vault



- WORM (Write Once Read Many) protection for backup data.
- Immutable retention period prevents modification or deletion.
- Strengthens protection against ransomware and insider threats.
- Meets regulatory and compliance requirements.

⑦ Best Practices



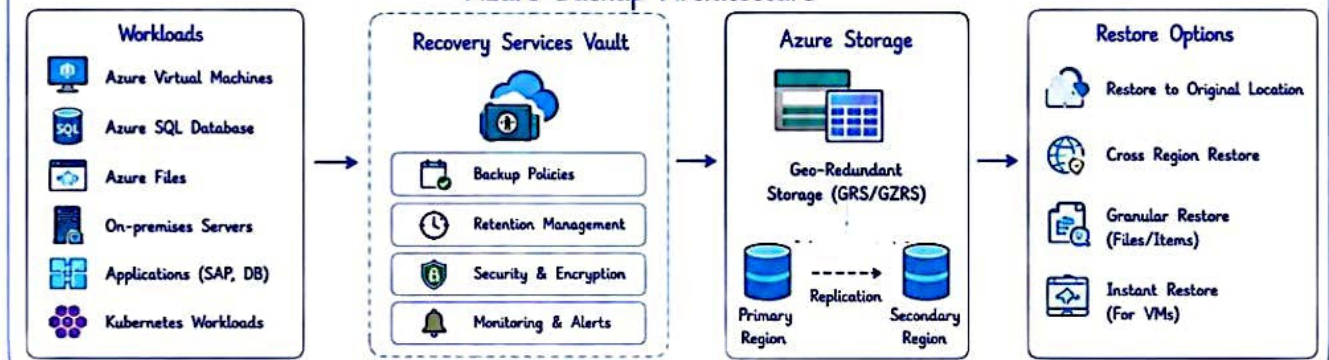
- Follow 3-2-1 Backup Rule.
- Encrypt backups at rest and in transit.
- Test recovery plan regularly.
- Monitor backup jobs and set up alerts.
- Use immutable vault for critical workloads.

⑧ Key Benefits

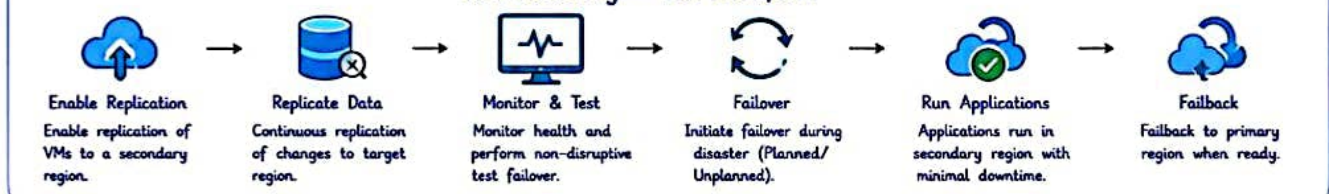


- High Availability
- Data Durability
- Business Continuity
- Ransomware Protection
- Cost Effective

Azure Backup Architecture



Site Recovery - DR Workflow



Best Practices Checklist

- ✓ Use Recovery Services Vault for all critical workloads.
- ✓ Follow 3-2-1 backup strategy.
- ✓ Enable Soft Delete and Immutable Vault.
- ✓ Store backups in geo-redundant storage.
- ✓ Regularly test backup and DR (Runbooks).
- ✓ Monitor backup jobs and set up alerts.
- ✓ Document and review DR plan periodically.



Common Use Cases



- VM and SQL Database Backup
- Ransomware Protection
- Region Outage Recovery



- Application Migration
- Compliance & Audit
- Development/Test Restore



Remember Backup protects your data. Disaster Recovery protects your business.



CLOUD
ARCHITECTURE



DEVOPS
AUTOMATION



SECURITY &
COMPLIANCE



COST OPTIMIZATION
& FINOPS

DESIGN

OPTIMIZE

SECURE

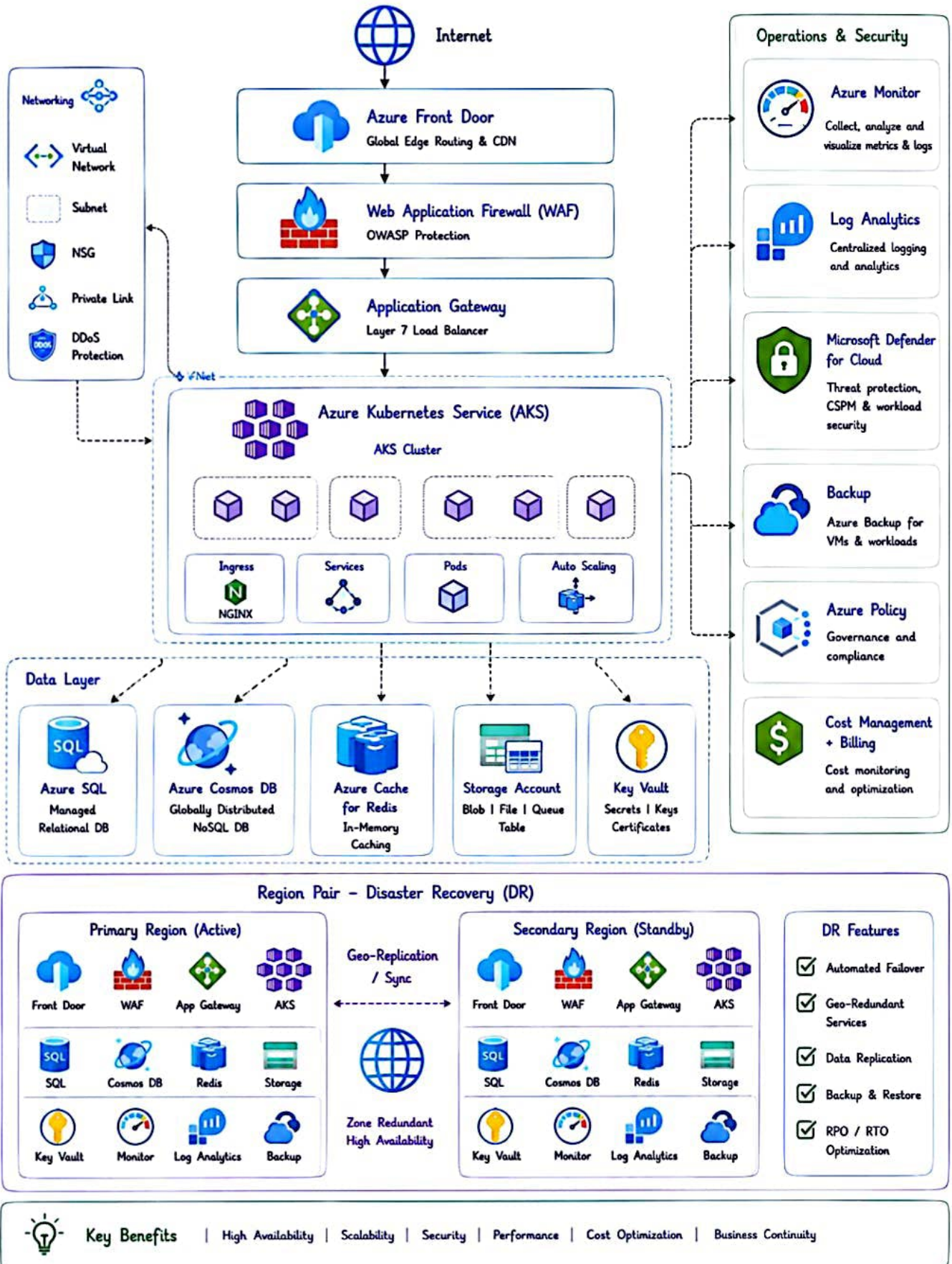
SCALE

Azure Architecture

Highly available, secure and scalable Azure production architecture with monitoring, security, backup and disaster recovery.



15





AZURE FIREWALL

Secure Your Cloud Network
with Intelligent Protection

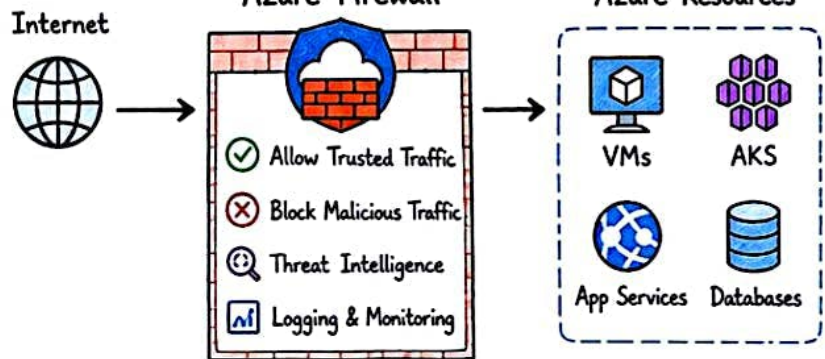


WHAT IS AZURE FIREWALL?

Azure Firewall is a managed, cloud-native network security service that protects Azure Virtual Networks.

It filters inbound, outbound, and east-west traffic using intelligent rules, threat intelligence, and centralized security policies.

HOW IT WORKS



WHAT CAN AZURE FIREWALL PROTECT AGAINST?

- ✓ Unauthorized Network Access
- ✓ Malicious IP Addresses
- ✓ Command & Control (C2) Traffic
- ✓ Port Scanning
- ✓ Lateral Movement
- ✓ Data Exfiltration
- ✓ Suspicious Outbound Connections
- ✓ Application Layer Threats (with Premium)



KEY FEATURES

- ☁ Fully Managed Service
- 🛡 Built-in High Availability
- 📊 Auto Scaling
- 🌐 Threat Intelligence Filtering
- 📄 Network Rules
- 🔗 Application Rules (FQDN Filtering)
- ↔️ DNAT & SNAT Support
- 📊 Azure Monitor & Log Analytics Integration
- 🛡 IDPS (Premium)
- 🔒 TLS Inspection (Premium)

DEPLOYMENT OPTIONS

★ STANDARD SKU ★

- ✓ Network Filtering
- ✓ Application Rules
- ✓ Threat Intelligence



★ PREMIUM SKU ★

- ✓ Everything in Standard
- ⊕ TLS Inspection
- ⊕ Intrusion Detection & Prevention (IDPS)
- ⊕ URL Filtering
- ⊕ Web Categories

BEST PRACTICES

- ✓ Enable Threat Intelligence Mode
- ✓ Use Firewall Policy for centralized management
- ✓ Integrate with Azure Monitor & Sentinel
- ✓ Enable Diagnostic Logs
- ✓ Use Application Rules over Network Rules where possible
- ✓ Restrict outbound Internet access
- ✓ Review logs regularly
- ✓ Follow Least Privilege principles



THINGS TO REMEMBER

- ✗ Azure Firewall is not a replacement for NSGs
- ✓ NSGs secure individual subnets/NICs
- ✓ Azure Firewall provides centralized traffic inspection
- ✓ Combine Azure Firewall + NSGs + Defender for Cloud for better security
- ✓ Monitor continuously and update firewall policies



DID YOU KNOW?

Azure Firewall can filter traffic at FQDN level, not just IP level! This helps protect against dynamic IPs and modern application threats.



KEY TAKEAWAY

Azure Firewall is your cloud network's security gatekeeper. It doesn't just allow or deny traffic—it provides intelligent inspection, centralized policy management, threat intelligence, and scalable protection for modern Azure environments.

Secure smarter. Monitor continuously. Stay resilient.



LET'S CONNECT!



Follow for more cloud security tips, diagrams and DevSecOps insights!

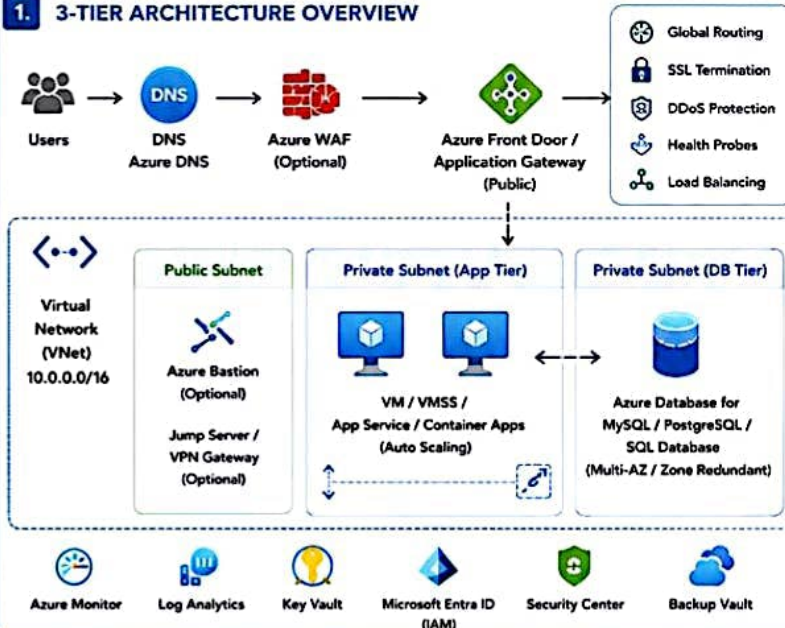




AZURE 3-TIER ARCHITECTURE CHEAT SHEET

HIGHLY AVAILABLE • SECURE • SCALABLE • COST OPTIMIZED

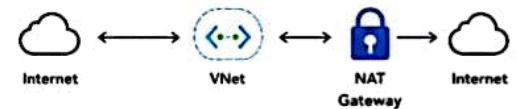
1. 3-TIER ARCHITECTURE OVERVIEW



2. VNET & NETWORKING

Subnet Type	CIDR (Example)	Purpose
Public Subnet	10.0.1.0/24	ALB/App Gateway, Bastion
Private Subnet (App)	10.0.2.0/24	Application / Web Tier
Private Subnet (DB)	10.0.3.0/24	Database Tier
Gateway Subnet	10.0.255.0/27	AzureBastionSubnet / GatewaySubnet

ROUTE TABLES	
Public Subnet Route Table	Private Subnet Route Table
0.0.0.0/0 → Internet	0.0.0.0/0 → NAT Gateway



3. LOAD BALANCING

- Use Azure Application Gateway (L7) for HTTP/HTTPS traffic
- SSL Termination with SSL Certificate (Azure Key Vault)
- Supports: Path-based routing, Host-based routing, WebSockets, Session Affinity
- Health Probe Path: /health
- WAF Protection (OWASP Top 10)

Listeners

- 443 (HTTPS) → Backend Pool
- 80 (HTTP) → Redirect to HTTPS

4. COMPUTE (APP TIER)



- Deploy in Private Subnet
- Use Availability Zones / Availability Sets
- Enable Auto Scaling based on CPU / Memory / Requests

5. DATABASE (DB TIER)



- Deploy in Private Subnet
- Enable Multi-AZ / Zone Redundant
- Enable Automated Backup
- Enable Private Endpoint (No Public Access)

6. SECURITY GROUPS (NSG)

App Subnet NSG	DB Subnet NSG
Inbound 80, 443 from App Gateway 22 from Bastion (optional) Outbound 3306 / 5432 / 1433 → DB Subnet 443 → Internet (via NAT)	Inbound 3306 / 5432 / 1433 from App Subnet Outbound - AllowVNetInbound - DenyAllOutbound

7. NETWORK SECURITY



8. IAM & ACCESS

- Use Microsoft Entra ID (Azure AD)
- Use Managed Identities (System/User Assigned)
- Role-Based Access Control (RBAC)
- Least Privilege Principle
- Disable Local Admin on VMs
- Use Azure RBAC, Not Access Keys

9. SECRETS & CERTIFICATES



10. STORAGE (AZURE STORAGE)

- Store Static Files (Images, CSS, JS)
- Enable Blob Encryption (SSE)
- Enable Versioning
- Lifecycle Management
- Block Public Access

11. MONITORING & ALERTING



12. LOGGING & AUDIT

- Azure Activity Log
- Azure Monitor Logs
- VNet Flow Logs
- Application Gateway Access Logs
- Azure SQL Auditing
- Store logs in Log Analytics / Storage Account

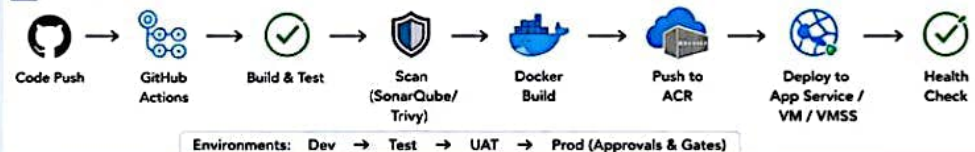
13. HIGH AVAILABILITY

- Multi-AZ / Zone Redundant Services
- Application Gateway with Zones
- SQL Database with Auto Failover
- VMSS with Zones
- Front Door for Global Availability
- Traffic Manager (DNS-based LB)

14. BACKUP & DISASTER RECOVERY

- Use Azure Backup for VMs
- Geo-Redundant Storage (GRS)
- SQL PITR (Point-in-time Restore)
- Cross Region Backup
- DR with Azure Site Recovery (ASR)

15. CI/CD PIPELINE (EXAMPLE)



16. USEFUL AZURE SERVICES

- | | |
|----------------------|-------------------------------|
| Azure Bastion | Secure RDP/SSH (No Public IP) |
| Azure Front Door | Global LB + CDN + WAF |
| Azure Key Vault | Secrets, Keys, Certificates |
| Azure Monitor | Monitoring & Alerts |
| Application Insights | Application Performance |
| Azure Policy | Governance & Compliance |
| Cost Management | Cost Analysis & Budgets |

17. BEST PRACTICES

- Keep Database Private (No Public Access)
- Use Private Endpoints
- Use Managed Identities
- Enable MFA for all users
- Encrypt Data at Rest (SSE, TDE)
- Encrypt Data in Transit (TLS 1.2+)
- Regular Patching & Updates
- Enable WAF & DDoS Protection
- Enable Auto Scaling
- Use RBAC & Least Privilege
- Store Secrets in Key Vault
- Backup Regularly
- Monitor Everything
- Implement CI/CD
- Follow Azure Well-Architected Framework

18. COMMON USE CASES

- Web Applications
- E-commerce Platforms
- SaaS Applications
- Enterprise Applications
- API Backends
- High Traffic Applications

AZURE VIRTUAL NETWORK PEERING

How Azure VNets Communicate Securely
Without the Internet



① WHAT IS VNET PEERING?

VNet Peering allows two Virtual Networks to connect to each other privately using the Microsoft backbone network.

- ✓ No VPN Gateway
- ✓ No Public IP
- ✓ Low Latency
- ✓ High Throughput
- ✓ Secure & Private Communication



② WHY NOT USE VPN?

VPN	VNET PEERING
✗ Uses Internet	✓ Uses Microsoft Backbone
✗ Higher Latency	✓ Low Latency
✗ Lower Throughput	✓ High Throughput
✗ Complex Setup	✓ Simple & Fast
✗ More Expensive	✓ Cost Effective

Use VNet Peering for VNets in Azure.
Use VPN for On-premises to Azure connectivity.

③ PEERING TYPES

1) LOCAL PEERING

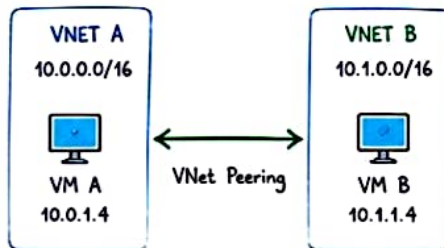
- Within the same region
- Lower latency
- Higher bandwidth



2) GLOBAL PEERING

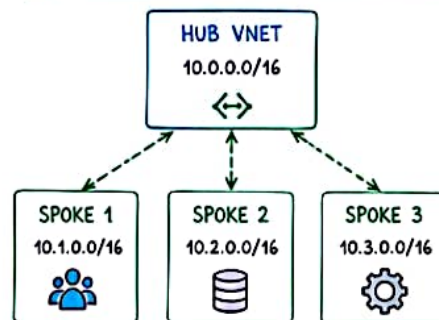
- Between VNets in different regions
- Uses Microsoft global backbone
- Slightly higher latency

④ HOW TRAFFIC FLOWS?



Traffic stays on Microsoft Backbone Network
(No Internet Involved)

⑤ HUB-AND-SPOKE ARCHITECTURE



Centralized connectivity, shared services,
and better management.

⑥ PEERING LIMITATIONS

- ✗ No Transitive Peering (A cannot reach C via B)
- ✗ Cannot Peer a VNet to itself
- ✗ Overlapping Address spaces are not allowed
- ✗ Peering is not supported across Azure Clouds
- ✗ No support for Gateway Transit



⑦ PRICING BASICS



- VNet Peering is FREE
- Data transfer charges apply (same as regular outbound/inbound rates)
- No additional gateway cost

⑧ REAL-WORLD USE CASE



Company has:

- Production VNet
- Development VNet
- Shared Services VNet (DNS, AD, Bastion)

Using VNet Peering, all VNets can securely communicate without exposing anything to the Internet.

⑨ COMMON MISTAKES

- ✗ Assuming Peering is Transitive
- ✗ Forgetting to enable Peering from both sides
- ✗ Overlapping Address Spaces
- ✗ Not planning IP Range Properly
- ✗ Ignoring NSG / UDR rules



⑩ INTERVIEW QUESTIONS

- Q1. What is VNet Peering?
- Q2. What is the difference between Local Peering and Global Peering?
- Q3. Can a VNet be peered with itself?
- Q4. Is VNet Peering transitive?
- Q5. What are the limitations of VNet Peering?
- Q6. When would you use VPN instead of VNet Peering?



⑪ KEY TAKEAWAYS



- Private & Secure communication
- High Performance with low latency
- Simple, Fast & Cost Effective
- Foundation for Enterprise Architectures

Think Private,
Think Peering!



⑫ NEXT LAB

Hands-on Lab:

- Create Two VNets
- Configure VNet Peering
- Deploy Two VMs
- Verify Communication (Ping Test)



PRO TIP



Always plan your IP address space and network architecture before creating peering. Good design = Secure & Scalable Cloud!



Keep Learning.
Keep Building.



Forward Proxy vs Reverse Proxy

Two Proxies. Different Jobs. Same Goal: Smarter Traffic Control.

Forward Proxy

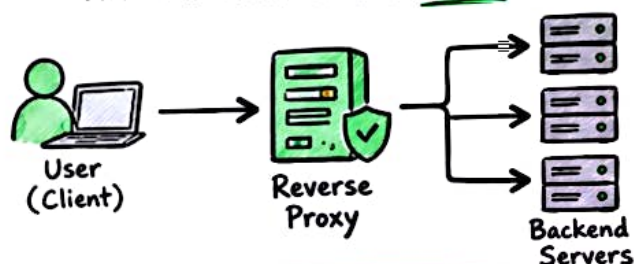
Acts on behalf of the client.



- ✓ Sits between client and internet
- ✓ Hides client identity
- ✓ Controls & filters outbound requests

Reverse Proxy

Acts on behalf of the server.



- ✓ Sits between internet and servers
- ✓ Hides server details
- ✓ Improves security, caching, load balancing.

Quick Comparison	Forward Proxy (Client-Side)	Reverse Proxy (Server-Side)
🎯 Purpose	Protect & filter client requests	Protect & optimize backend services
📍 Location	Between <u>Client</u> & <u>Internet</u>	Between <u>Internet</u> & <u>Server</u>
👤 Who Initiates	Client initiates the request	Server receives the request
👁️ Visibility	Hides client IP	Hides server IP & architecture
★ Examples	Corporate Proxy, Squid, Zscaler	Nginx, HAProxy, AWS ALB, Cloudflare

Use Cases

✓ Forward Proxy:

- Employee internet access control
- Bypassing geo-restrictions
- Caching content to save bandwidth
- Monitoring & logging user activity



Use Cases

Reverse Proxy:

- Load balancing across servers
- SSL termination
- Caching static content
- WAF & DDoS protection



Key Takeaway: Forward Proxy = For the Client | Reverse Proxy = For the Server
 "Different position. Same mission: Security, Performance & Efficiency."



Azure Private Endpoints vs Service Endpoints

Both improve security, but in different ways!



Private Endpoints

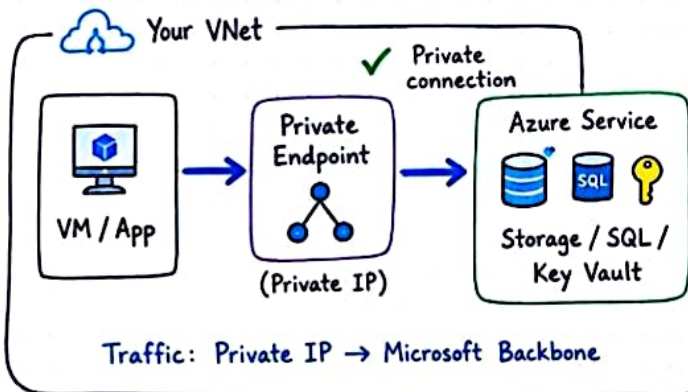
- Uses Private IP from your VNet
- Traffic stays on Microsoft backbone
- Resource looks like it's in your VNet
- No public endpoint exposure



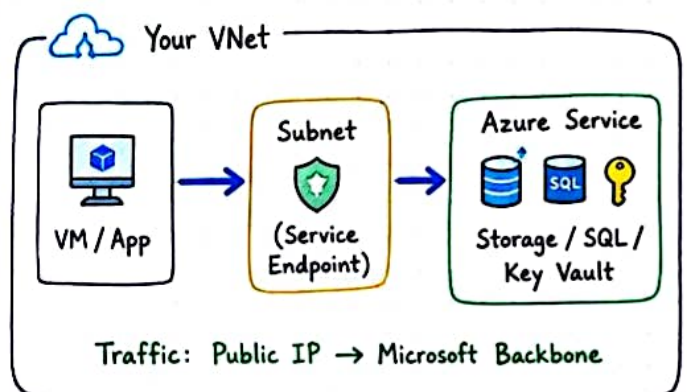
Service Endpoints

- Uses Microsoft backbone
- Still uses public IP of the service
- Scoped to specific Azure services
- Easier and cost-effective

1 Private Endpoints - How it works



2 Service Endpoints - How it works



3 Quick Comparison

Aspect	Private Endpoints	Service Endpoints
🎯 Connectivity	Private IP in your VNet	Public IP of the service
🔍 Scope	Resource-level (one by one)	Subnet-level (multiple resources)
🛡️ Access Control	NSG + Private DNS	NSG + Service Endpoint Policy
👁️ Public Exposure	No	Yes (public endpoint exists)
💰 Cost	Billed per endpoint	Free
★ Best For	High security, compliance, isolation	Simplicity, scale, cost-efficiency

4 When to use what?

Use Private Endpoints when:

- ✓ You handle sensitive data
- ✓ You need strict security & compliance
- ✓ You want zero public exposure
- ✓ Example: Finance, Healthcare, Prod workloads

Use Service Endpoints when:

- ✓ You want simple and quick setup
- ✓ You need access to multiple services
- ✓ Cost is a key factor
- ✓ Example: Dev/Test, Non-sensitive workloads

5 Services that support Service Endpoints



Storage Account



SQL Database



Key Vault



Azure Cosmos DB

...

★ Pro Tip

You can use both together in a strong networking strategy for layered security and performance. ✓

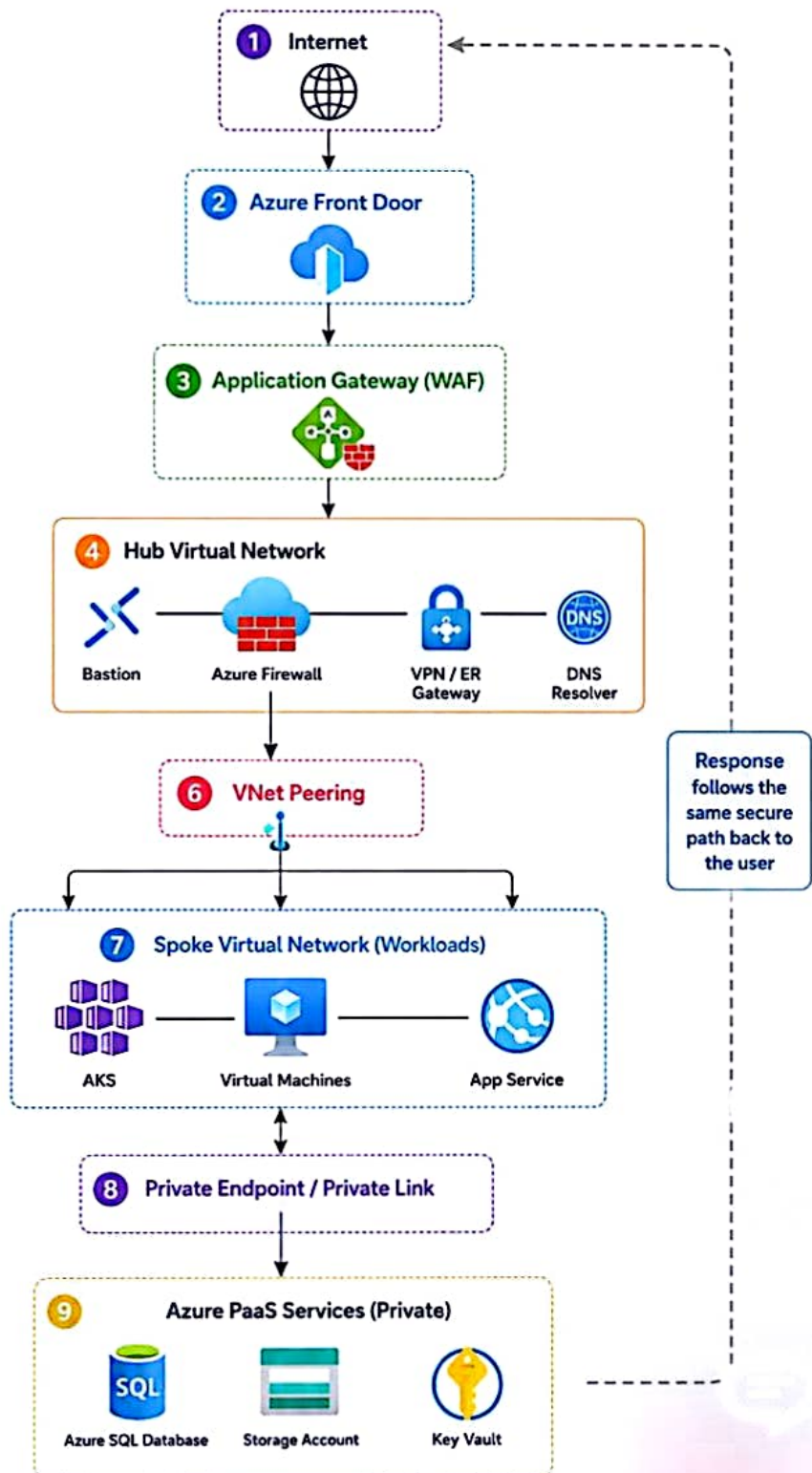
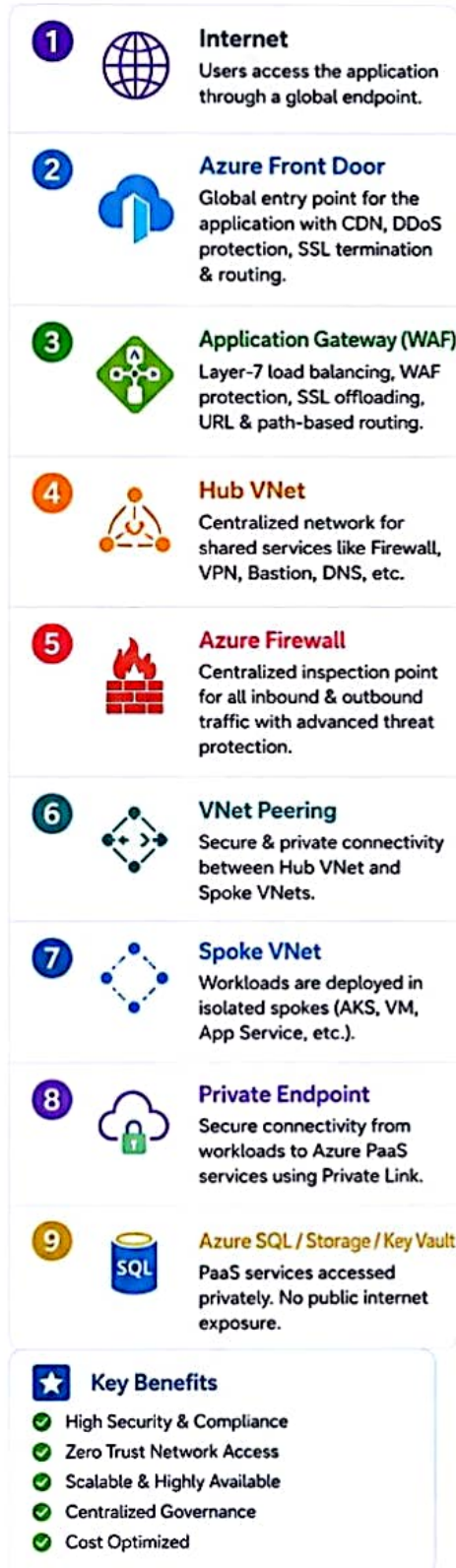


Azure Landing Zone Traffic Flow

Secure. Scalable. Well-Governed.



A modern enterprise-ready architecture using Azure Hub & Spoke model with best practices for security, performance & scalability.



Architecture Highlights



Security First

Traffic inspected at Azure Firewall & WAF with Private Link connectivity.



Global & Scalable

Front Door provides global reach, load balancing & high availability.



Hub & Spoke Model

Centralized security & shared services in Hub with isolated Spoke workloads.



Governance

Aligned with Azure Landing Zone best practices & policies.



AZURE LANDING ZONE

An Azure Landing Zone is a pre-configured, secure and scalable Azure environment that provides the foundation for deploying workloads. It follows Microsoft's best practices for governance, security, networking, identity and resource organization.

Foundation
for your
cloud journey
✓

1 CORE COMPONENTS



Management Groups
Organize subscriptions



Subscriptions
Separate workloads or environments



Resource Groups
Group related Azure resources



Identity & Access
Microsoft Entra ID (Azure AD) and RBAC



Networking
VNets, NSGs, Firewalls, Hub-Spoke architecture



Governance
Azure Policy, Blueprints, Tags



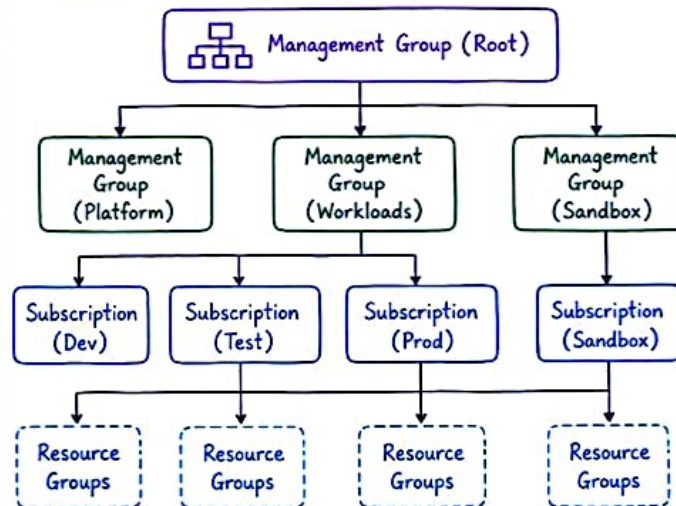
Monitoring
Azure Monitor, Log Analytics

3 BENEFITS

- ✓ Secure by default
- ✓ Standardized architecture
- ✓ Easy governance and compliance
- ✓ Scalable for enterprise growth
- ✓ Faster cloud deployment



2 LANDING ZONE ARCHITECTURE (EXAMPLE)



Built-in Best Practices

- ✓ Security
- ✓ Governance
- ✓ Compliance
- ✓ Scalability
- ✓ Cost Optimization
- ✓ Operational Excellence

4 REAL-LIFE EXAMPLE



A company wants separate environments for Development, Testing and Production. Instead of creating everything manually each time, it builds an Azure Landing Zone once with security, networking, policies and monitoring already configured.

Every new application is deployed into this ready-made, secure foundation.

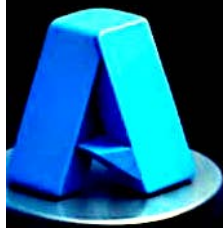
5 ONE-LINE INTERVIEW ANSWER

“An Azure Landing Zone is a pre-configured, secure and scalable Azure environment that provides a standardized foundation for deploying and managing cloud workloads using Microsoft's best practices.”



KEY TAKEAWAY

Azure Landing Zone = Foundation + Governance + Security + Networking + Monitoring for a successful cloud adoption 😊



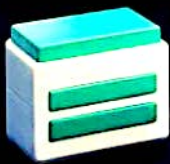
AZURE STORAGE & DATABASE

Store. Secure. Scale.
Powering Modern Applications in the Cloud



STORAGE SERVICES

Scalable, Durable & Secure Storage
for Every Need



1 Azure Storage Account

A unified storage solution that provides access to multiple Azure Storage services.



2 Blob Storage

Store massive amounts of unstructured data like images, videos, backups & logs.



3 Azure Files

Fully managed file shares accessible via SMB or NFS protocol.



4 Queue Storage

Reliable message queueing for asynchronous communication between apps.



5 Table Storage

NoSQL key-value store for schemaless, structured data with massive scalability.



6 Managed Disks

High-performance persistent storage for Azure Virtual Machines.



DATABASE SERVICES

Powerful, Managed & Intelligent Databases



7 Azure SQL Database

Fully managed relational database with automatic scaling, backups & high availability.



8 Azure SQL Managed Instance

Lift & shift existing SQL Server workloads with near 100% compatibility.



9 Azure Cosmos DB

Globally distributed NoSQL database with multi-region replication & ultra-low latency.



10 Azure Database for PostgreSQL

Fully managed PostgreSQL database for modern cloud applications.



11 Azure Database for MySQL

Fully managed MySQL database with automated maintenance & backups.



12 Azure Cache for Redis

In-memory data store that boosts performance & reduces database workload.

KEY TAKEAWAY

Choose the right service based on your workload:



Files & Documents
→ Azure Files



Images, Videos & Backups
→ Blob Storage



VM Storage
→ Managed Disks



Relational Apps
→ Azure SQL DB /
SQL Managed Instance



NoSQL & Global Scale
→ Cosmos DB



Open-Source Databases
→ PostgreSQL /
MySQL



High-Speed Caching
→ Azure Cache
for Redis



REMEMBER:

The right storage or database service ensures high availability, security, and cost optimization for your cloud applications.

Keep Learning. Keep Building! 🚀





AZURE WAF & SSL

Powerful Security. Encrypted Connections. Protected Applications.

Secure your web applications with Azure WAF and SSL/TLS

VIKRAL

The Guardian
Blocks Malicious
Traffic & Attacks

JABRAL

The Protector
Safeguards Applications
& Ensures Availability

USERS



INTERNET



AZURE WAF
Web Application Firewall



SSL/TLS

Encrypts data in transit
to ensure privacy and
data integrity.

KEY TAKEAWAY

Azure WAF filters and blocks malicious traffic,
while SSL/TLS encrypts data in transit.
Together, they deliver end-to-end security
for your applications.

SSL/TLS BENEFITS

- Encrypts data between client and server
- Protects against eavesdropping
- Ensures data integrity
- Builds user trust
- SSL Certificates from Azure Key Vault

BACKEND APPLICATIONS

- Web App 1
- Web App 2
- Web App 3



- SQL INJECTION
- XSS ATTACKS
- BOT TRAFFIC
- IP SPOOFING
- DDoS ATTACKS

WAF FEATURES

- OWASP Top 10 Protection
- Custom Rules & Rate Limiting
- IP & Geo Filtering
- Bot Protection
- Real-time Threat Intelligence
- Integration with Azure Monitor



Stronger Security



Data Encryption



High Performance



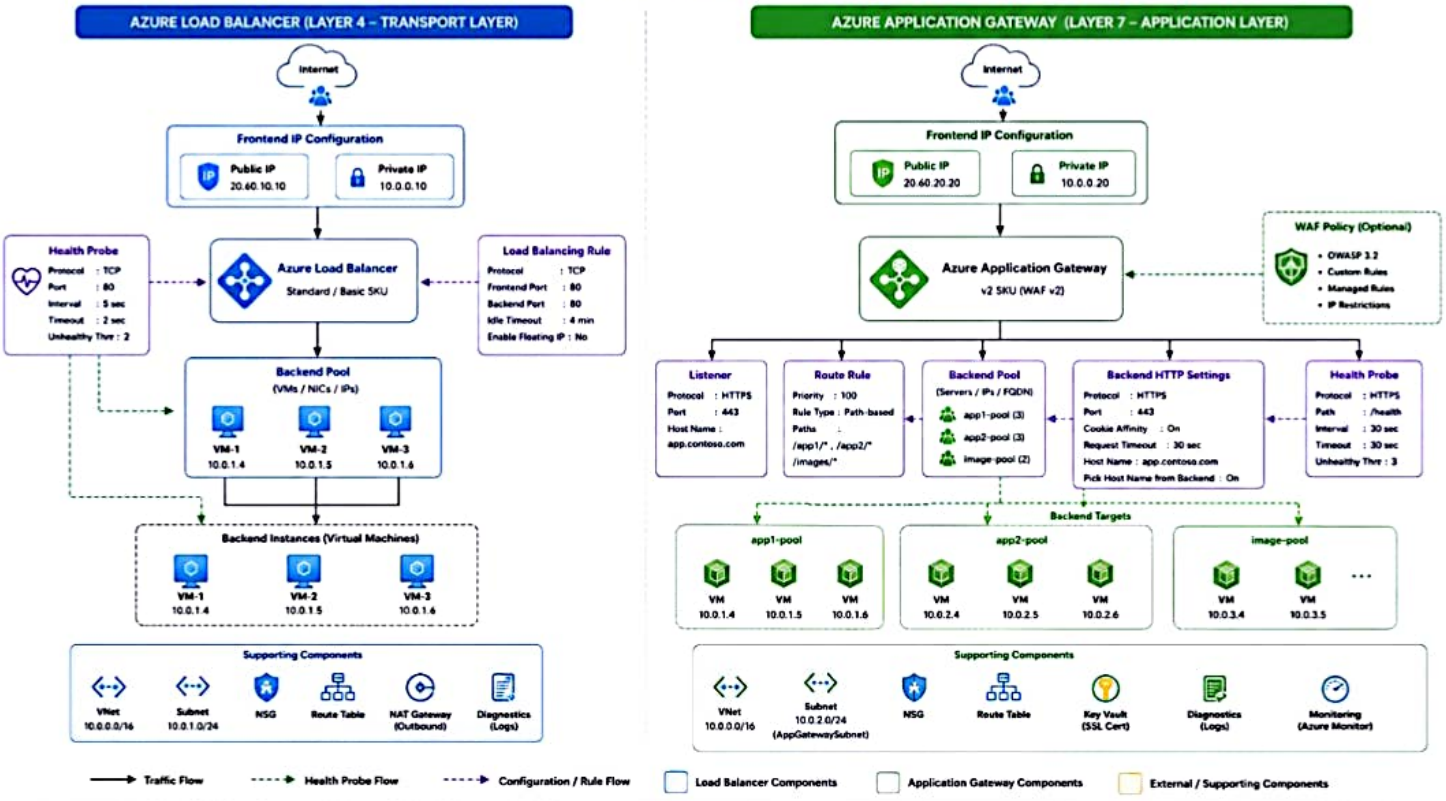
Scalable



Compliance Ready

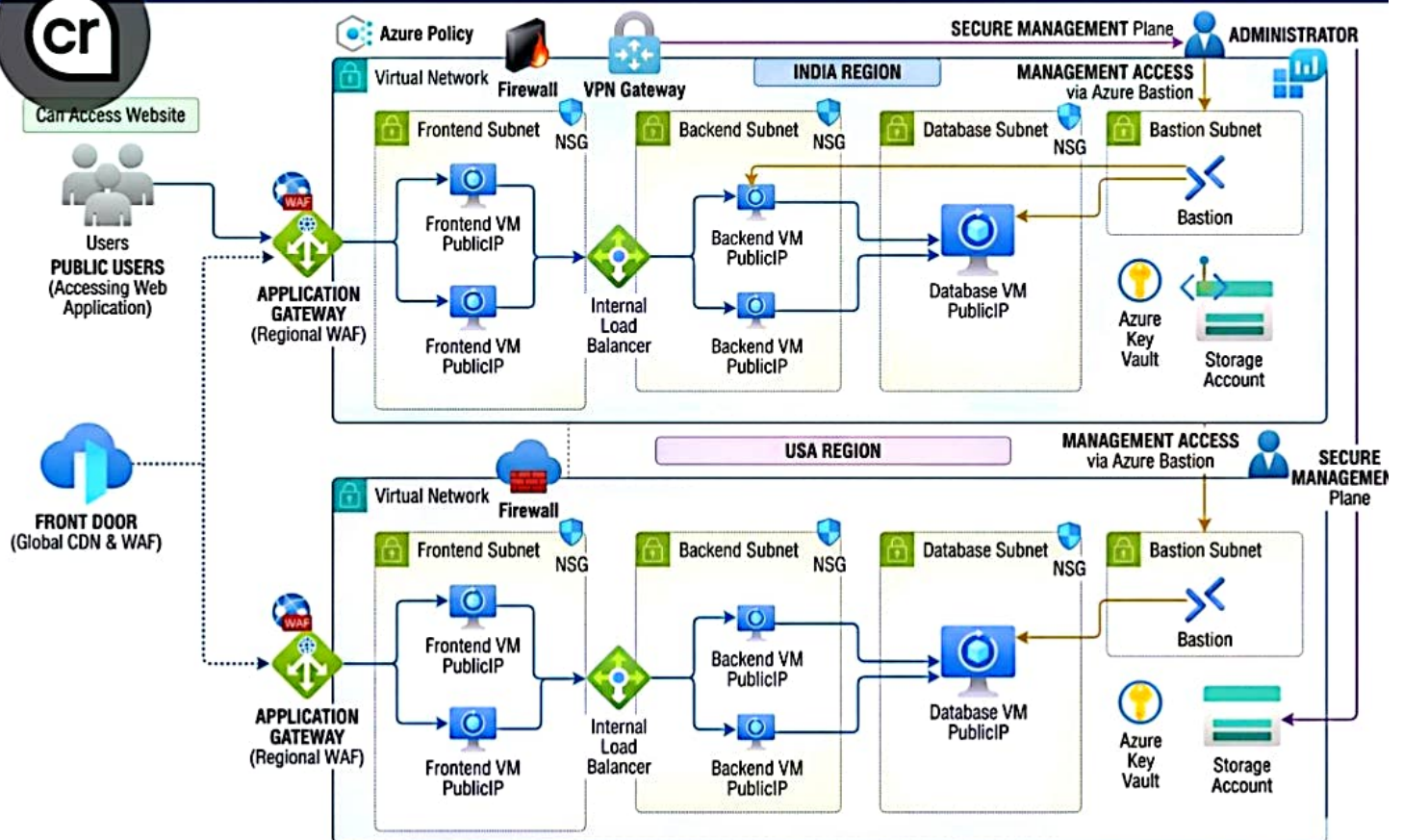
Azure Load Balancer vs Azure Application Gateway

Deep Flow & All Resources ●●●



AT A GLANCE

OSI LAYER	ROUTING BASED ON	PROTOCOLS	SESSION AWARE
Layer 4 (Transport Layer)	IP + Port	TCP, UDP	No
VS			
Layer 7 (Application Layer)	URL, Path, Host Header, Query String, Cookies	HTTP, HTTPS	Yes (Cookie Based Affinity)



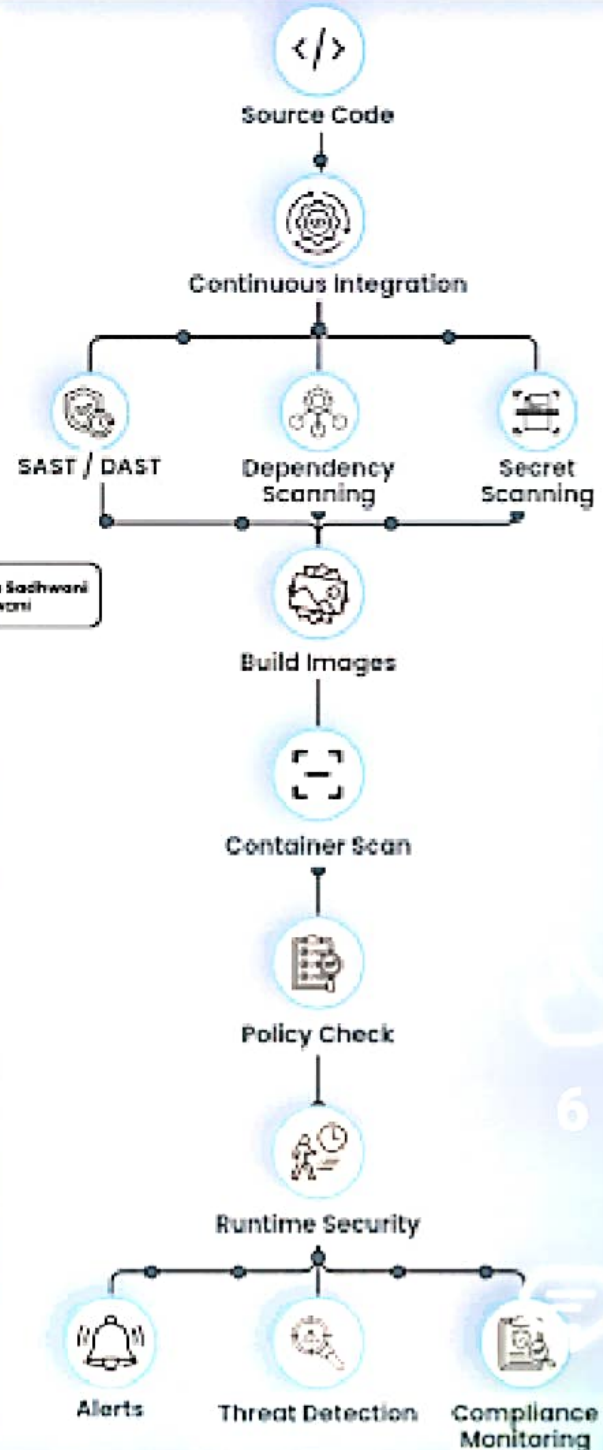
Traditional Security VS DevSecOps

Traditional Security



Traditional Security checks risk near the end.
Security as a gate

DevSecOps



DevSecOps checks risk throughout delivery.
Security inside CI/CD

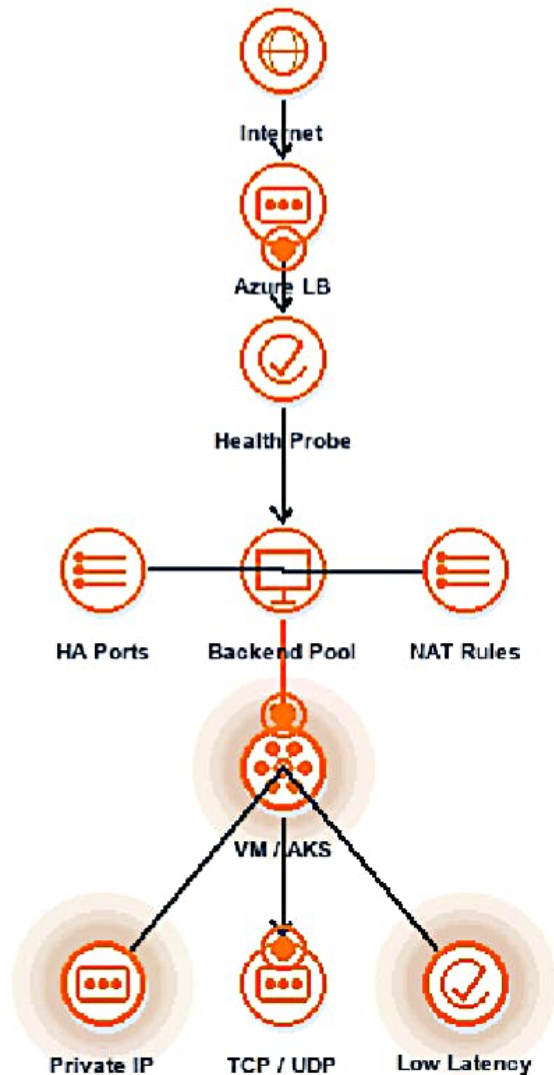
Vishakha Sachwani
@vrsachwani

Load Balancer VS App Gateway

Rahul Goyal

Azure Load Balancer

Layer 4 | TCP / UDP



Simple Rule

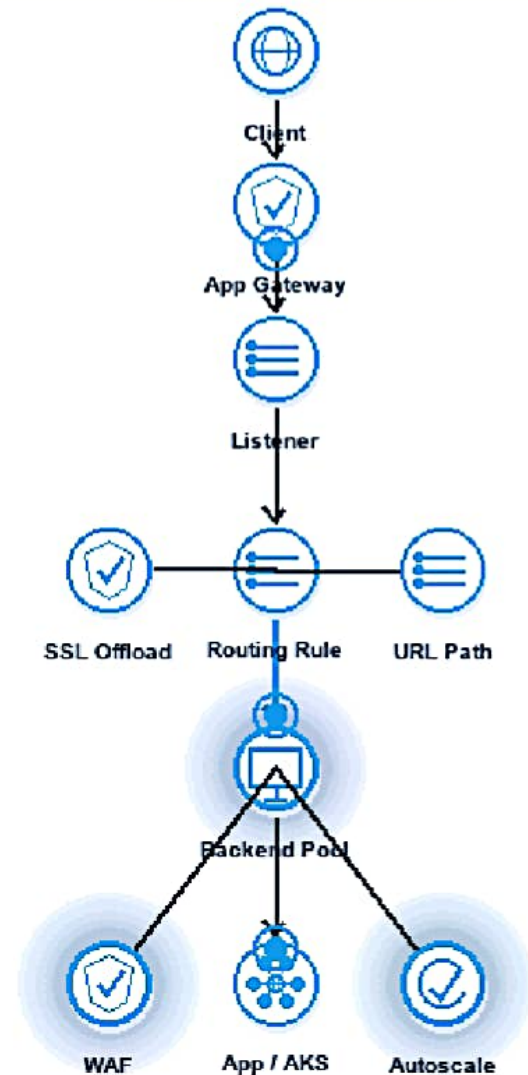
Network traffic

Load Balancer distributes network traffic.

Fast L4 traffic for VMs, AKS, HA ports.

Application Gateway

Layer 7 | HTTP / HTTPS



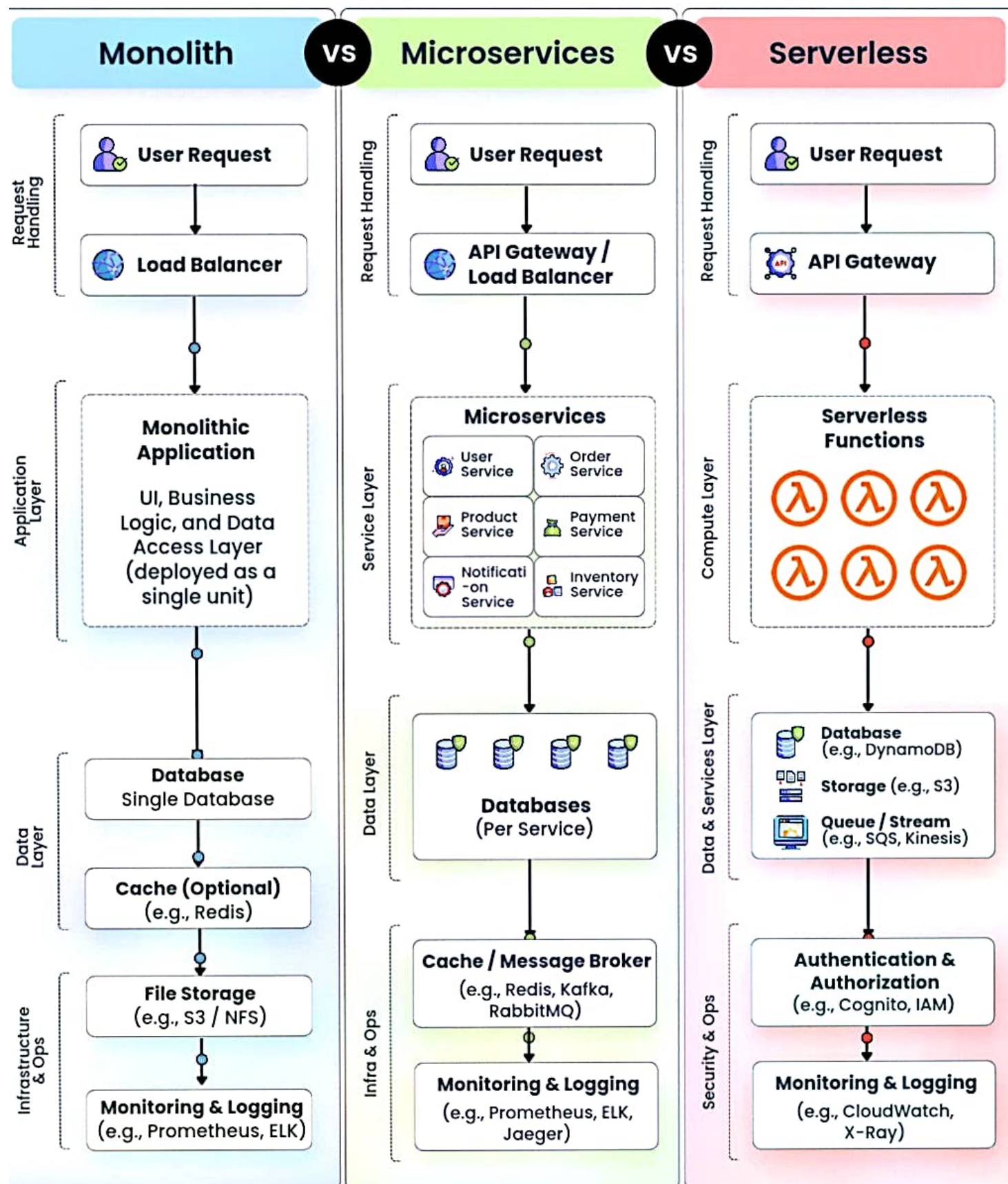
Simple Rule

Web traffic

Application Gateway routes web traffic.

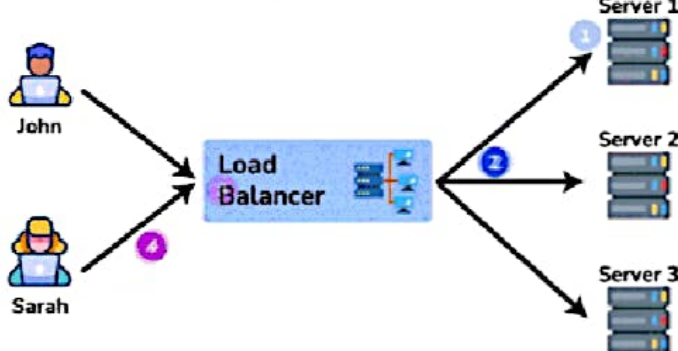
WAF, SSL offload, URL/path routing.

Monolith vs Microservices vs Serverless

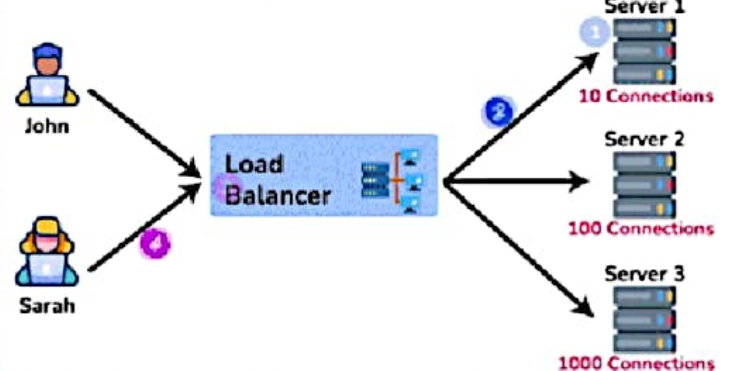


Load Balancing Algorithms

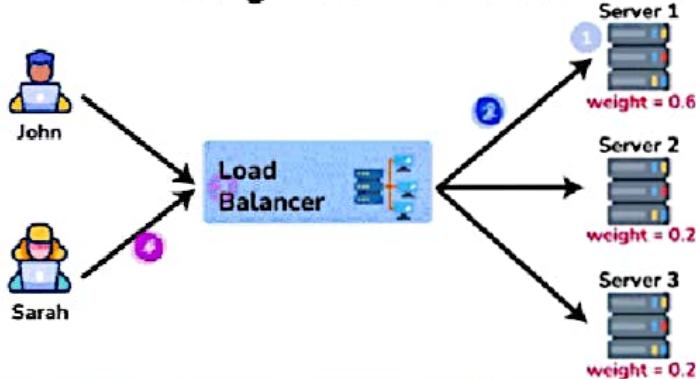
1. Round Robin



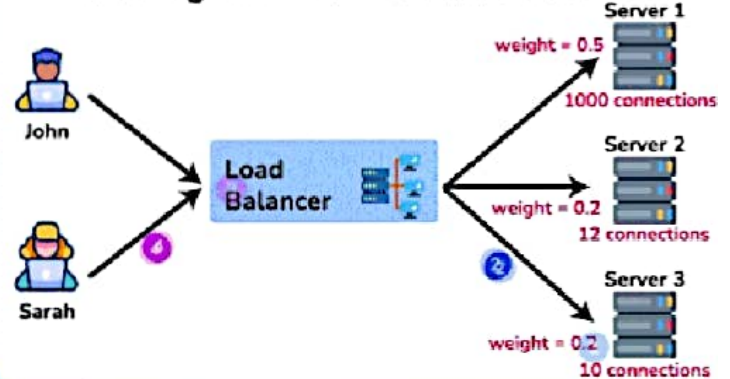
2. Least Connections



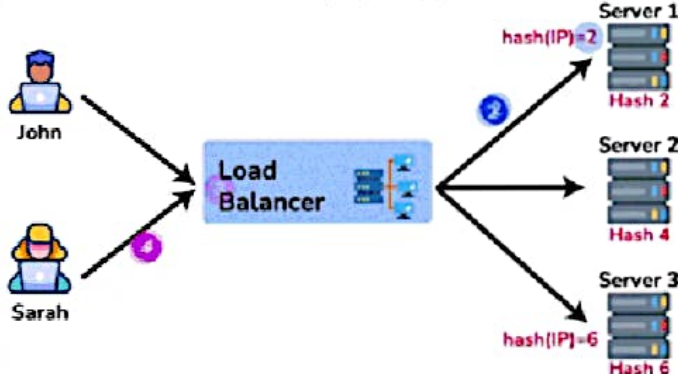
3. Weighted Round Robin



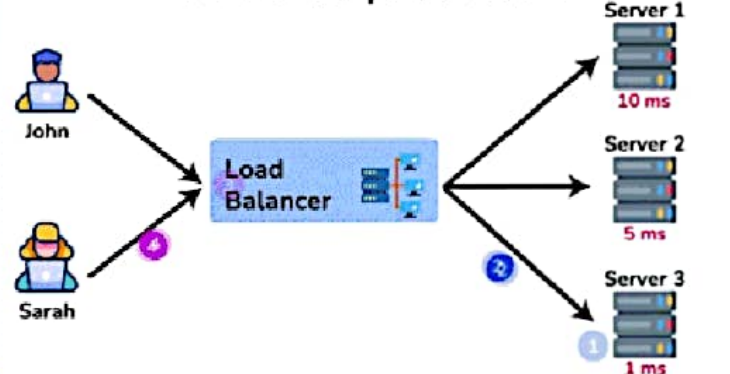
4. Weighted Least Connections



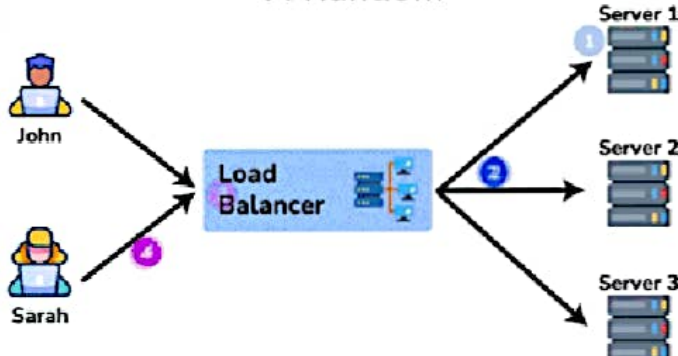
5. IP Hash



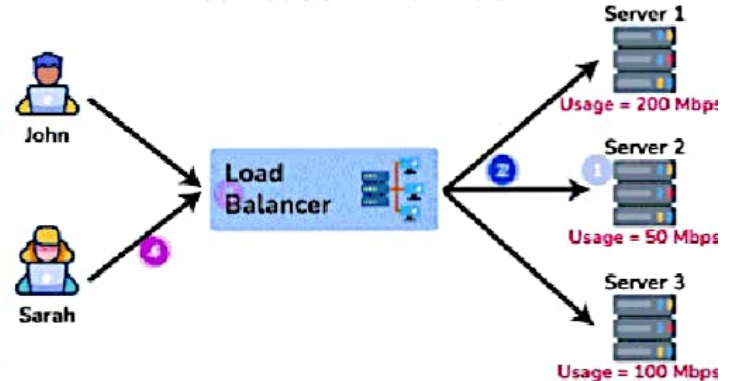
6. Least Response Time



7. Random



8. Least Bandwidth



STATIC IP · VS · DYNAMIC IP

A Simple Way to Understand the Difference



STATIC IP

Stays the same unless changed manually



Example IP Address
203.0.113.10

Commonly Used For



Servers



Websites



CCTV Systems



Applications

Think of it like:



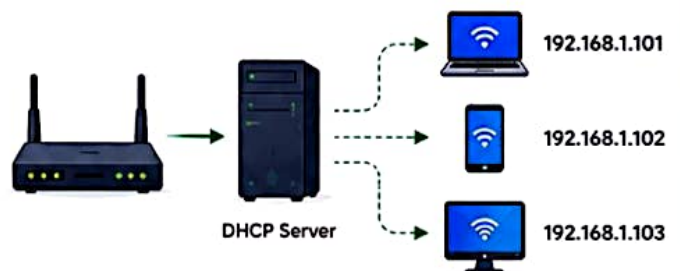
OWNING A HOUSE

You always have the same address.



DYNAMIC IP

Assigned automatically by DHCP server and can change over time



Commonly Used For



Home Internet



Everyday Browsing

Think of it like:



TAKING A TAXI

Your ride changes, but you still reach your destination.



There isn't a "better" option. Each serves a different purpose!



Use **Static IP** for hosting, servers, remote access.



Use **Dynamic IP** for everyday internet browsing.



Understanding these basics is a small step that builds a strong foundation for networking, cloud computing, and DevOps.



What analogy helped you understand the difference between **Static** and **Dynamic** IPs? I'd love to hear it!





Important Ports Used in DevOps



Stay Connected. Stay Automated. Stay Ahead!

SERVICE / TOOL	PORT	PROTOCOL	USE / DESCRIPTION
 SSH	22	TCP	Secure Shell Access
 HTTP	80	TCP	Web Traffic (Unsecured)
 HTTPS	443	TCP	Secure Web Traffic
 FTP	21	TCP (21) + (20)	File Transfer Protocol
 SFTP	22	TCP	Secure File Transfer
 Git	9418	TCP	Git Protocol
 GitHub	22, 443	TCP	Code Hosting & Git Access
 GitLab	22, 443	TCP	Code Hosting & DevOps Platform
 Jenkins	8080	TCP	CI/CD Automation Server
 Docker Registry	5000	TCP	Docker Image Registry
 Kubernetes API Server	6443	TCP	K8s API Communication
 Docker Swarm Management	2377, 2376	TCP	Swarm Cluster Management
 SonarQube	9000	TCP	Code Quality Analysis
 Nexus Repository	8081, 8082	TCP	Artifact / Package Repository
 Ansible	22	SSH (TCP)	Configuration Management
 Terraform (Remote Backend)	443	HTTPS (TCP)	Remote State Storage
 Prometheus	9090	TCP	Monitoring & Alerting
 Grafana	3000	TCP	Data Visualization
 ELK Stack (Elasticsearch)	9200	TCP	Search & Analytics Engine
 MySQL	3306	TCP	Database
 PostgreSQL	5432	TCP	Database
 Redis	6379	TCP	In-Memory Data Store
 RDP	3389	TCP	Remote Desktop Protocol (Windows Access)



Always ensure required ports are open in Firewall / Security Group



Code



Build



Test



Deploy



Monitor

★ Remember ★
Open the right port,
for the right service,
for the right access.
😊



DNS RECORD TYPES EXPLAINED

DNS is like the **Internet's Phonebook**.
It translates domain names into IP addresses and other information.



1 A RECORD (Address Record)



Maps a domain name to an
IPv4 address.

example.com → 192.168.1.10

EXAMPLE

example.com → 192.168.1.10

2 AAAA RECORD (Quad-A Record)



Maps a domain name to an
IPv6 address.

example.com → 2001:db8::1

EXAMPLE

example.com → 2001:db8::1

3 CNAME RECORD (Canonical Name)



Maps one domain name to
another **domain name**.

www.example.com → example.com

EXAMPLE

www.example.com → example.com

4 MX RECORD (Mail Exchange)



Specifies the **mail server** responsible
for receiving emails.

example.com → mail.google.com

EXAMPLE

example.com → mail.google.com

5 TXT RECORD (Text Record)



Stores **text-based information**
for verification and policies.

example.com → v=spf1 include:_spf.google.com ~all

EXAMPLE

example.com → "v=spf1 include:_spf.google.com ~all"

6 NS RECORD (Name Server)



Specifies the **authoritative name servers**
for the domain.

example.com → ns1.cloudflare.com
ns2.cloudflare.com

EXAMPLE

example.com → ns1.cloudflare.com

7 PTR RECORD (Pointer Record)



Maps an IP address to a
domain name (**Reverse DNS**).

192.168.1.10 → server.example.com

EXAMPLE

192.168.1.10 → server.example.com

8 SOA RECORD (Start of Authority)



Contains important information about
the **DNS zone**.

- Primary Name Server
- Admin Email
- Serial Number
- Refresh Interval
- Retry Interval
- Expire Time
- Minimum TTL

EXAMPLE

example.com → SOA Record Details



INTERVIEW CHEAT SHEET

A RECORD vs CNAME

A vs **CNAME**

A maps to an IP address.
CNAME maps to another
domain name.

MX vs TXT

MX vs **TXT**

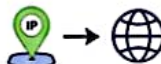
MX is used for email routing.
TXT is used for verification
and policies.

NS vs SOA

NS vs **SOA**

NS defines the name servers.
SOA contains zone
information and settings.

PTR RECORD



Used for Reverse DNS,
important for email servers
and security.

Why TXT is Important?



Helps in SPF, DKIM,
DMARC and domain
verification.